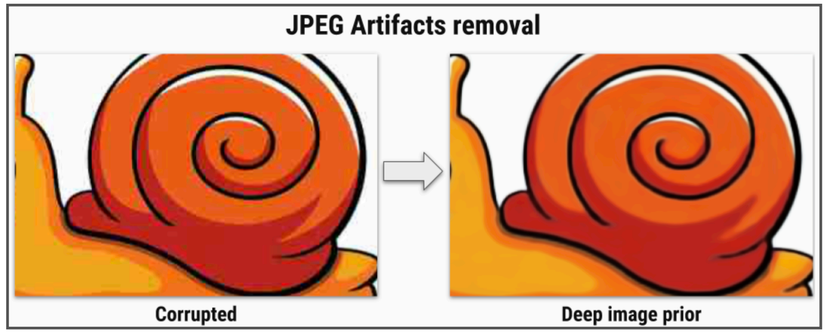
Neural Surface Reconstruction

February 14th, 2024

Previous lectures …

- Inverse problems

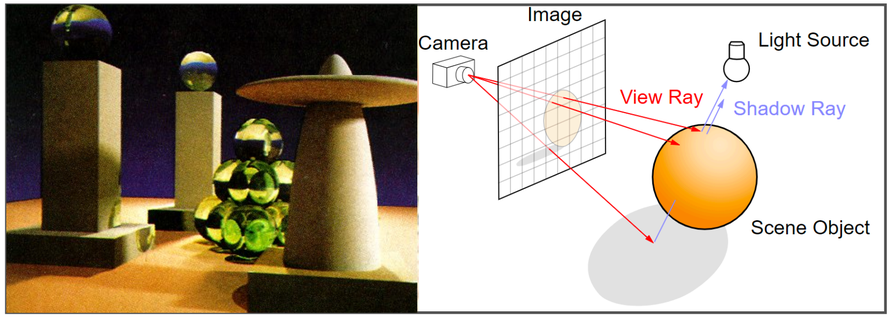


Deep Image Prior [Ulyanov et al. 2018]

- Structure from motion (2D to 3D)

Photo Tourism: [Snavely et al. 2006]

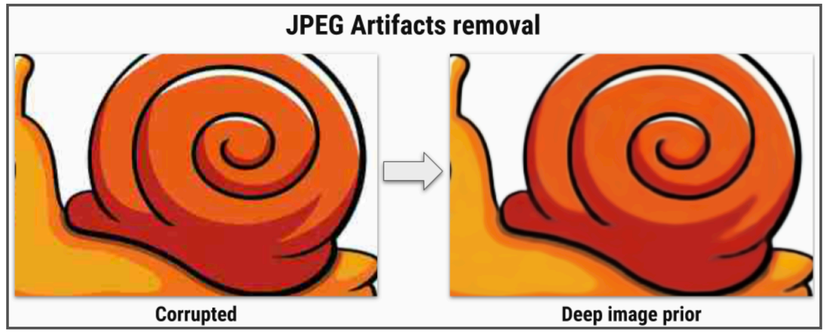
- Rendering (3D to 2D)



The Rendering Equation [Kajiya 1986]

Previous lectures …

- Inverse problems

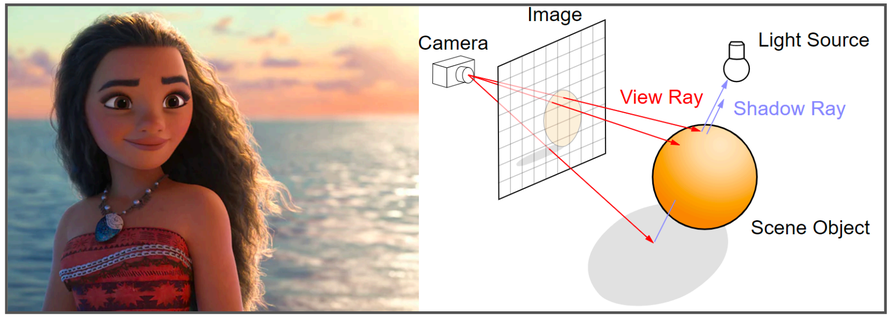


Deep Image Prior [Ulyanov et al. 2018]

- Structure from motion (2D to 3D)

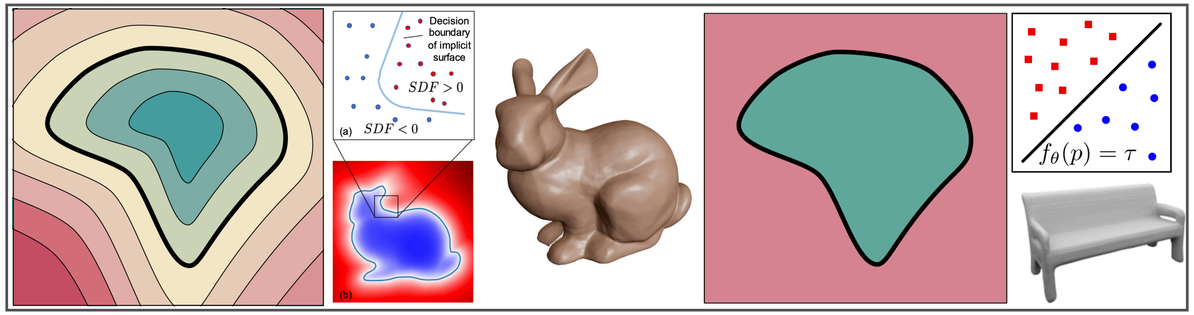
Photo Tourism: [Snavely et al. 2006]

- Rendering (3D to 2D)



The Rendering Equation [Kajiya 1986]

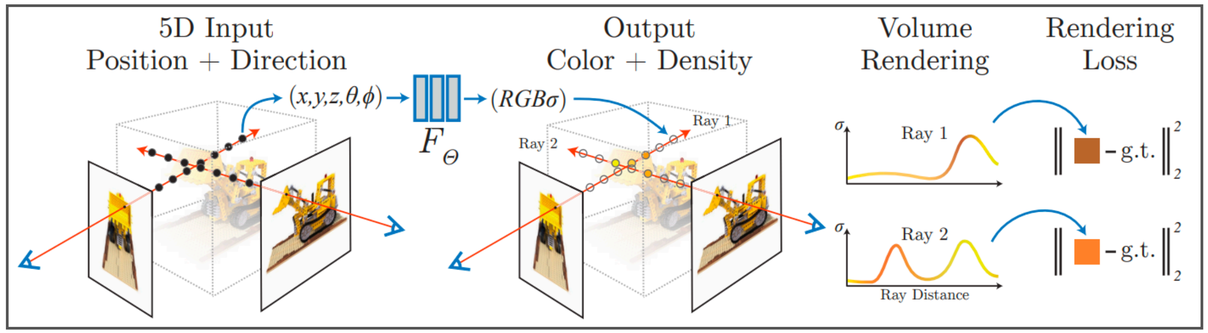
- Implicit Neual Representations



DeepSDF [Park et al. 2019]

Occupancy Networks Mescheder et al. 2019]

- Neural Rendering

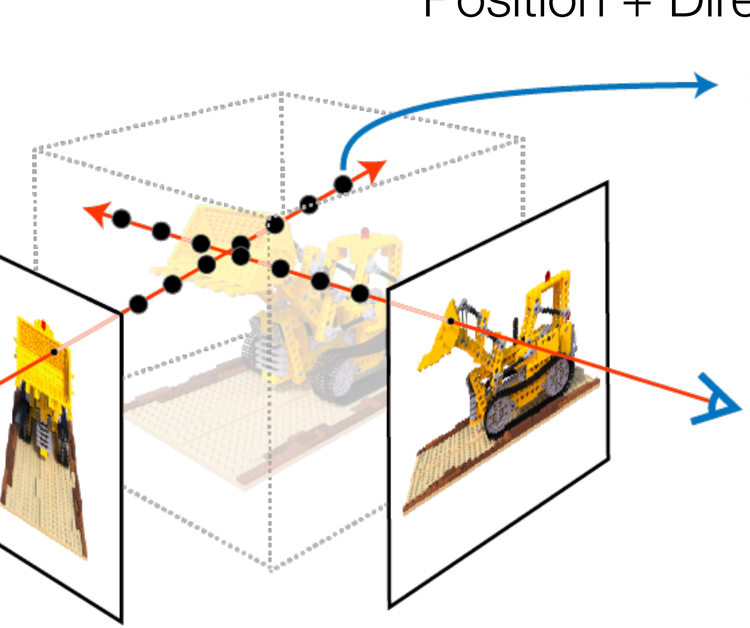


Neural Radiance Fields [Mildenhall et al. '20]

Neural Radiance Fields (NeRF)

Input:
Position + Direction

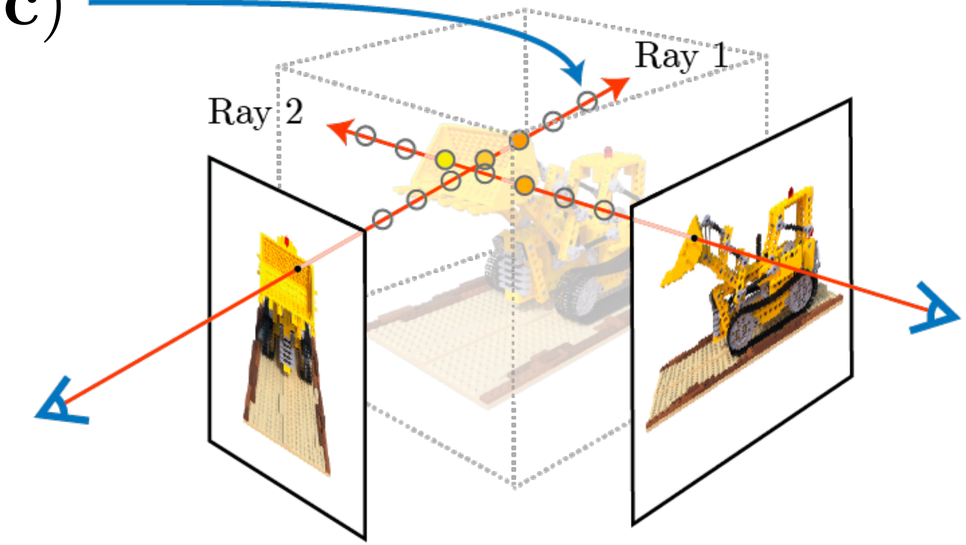
$(\mathbf{x}, \hat{\mathbf{d}})$



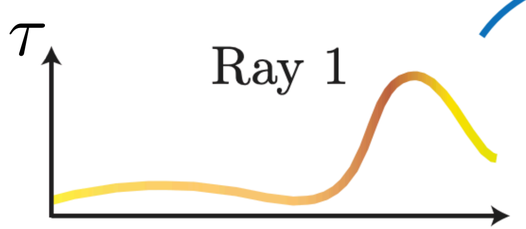
F_Θ

Output:
Density + Color

$(\tau, \mathbf{c})$



Volume rendering



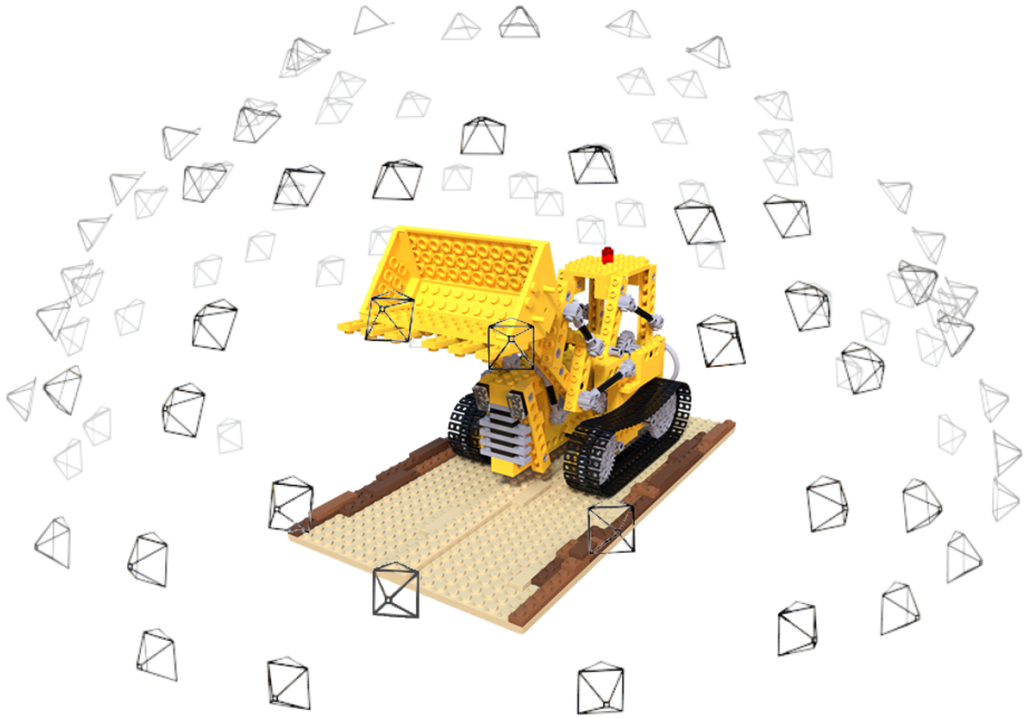
Rendering loss

$$\left\| \blacksquare - \text{g.t.} \right\|_2^2$$

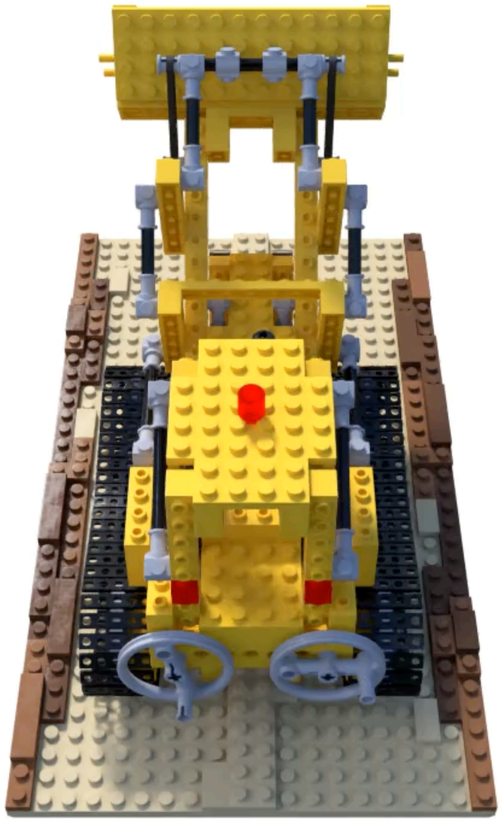
NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis, Mildenhall, Srinivasan, Tancik, et al., 2020

Neural Radiance Fields (NeRF)

Input images + cameras



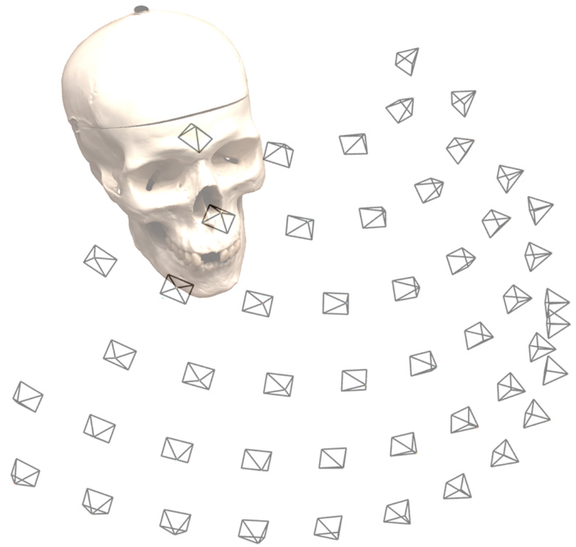
Novel views synthesis



NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis, Mildenhall, Srinivasan, Tancik, et al., 2020

Today's lecture

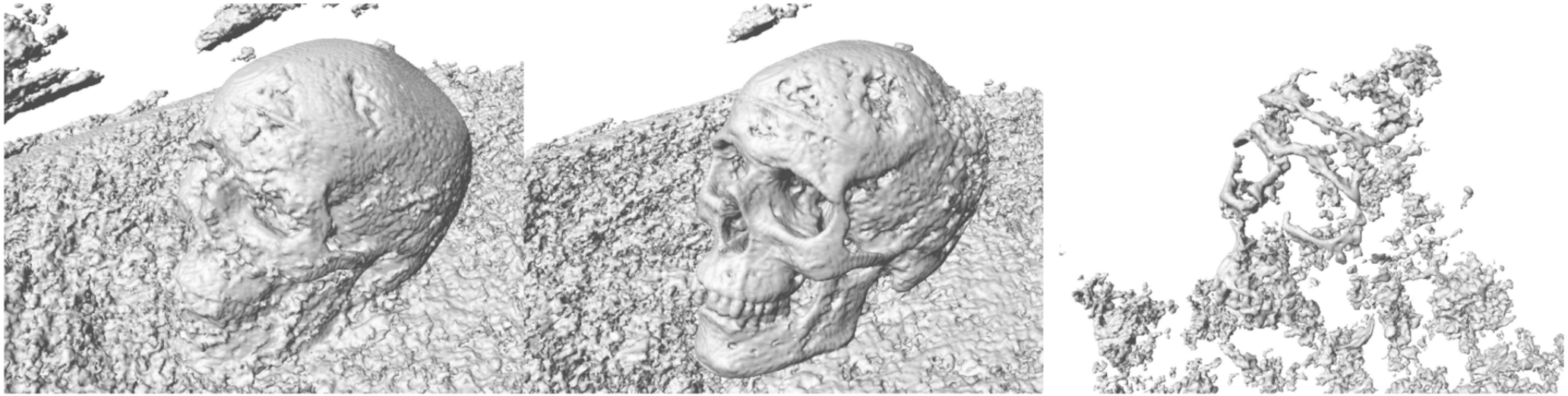
Multiview images



Camera poses

NeRF - Differential **Volume** Rendering

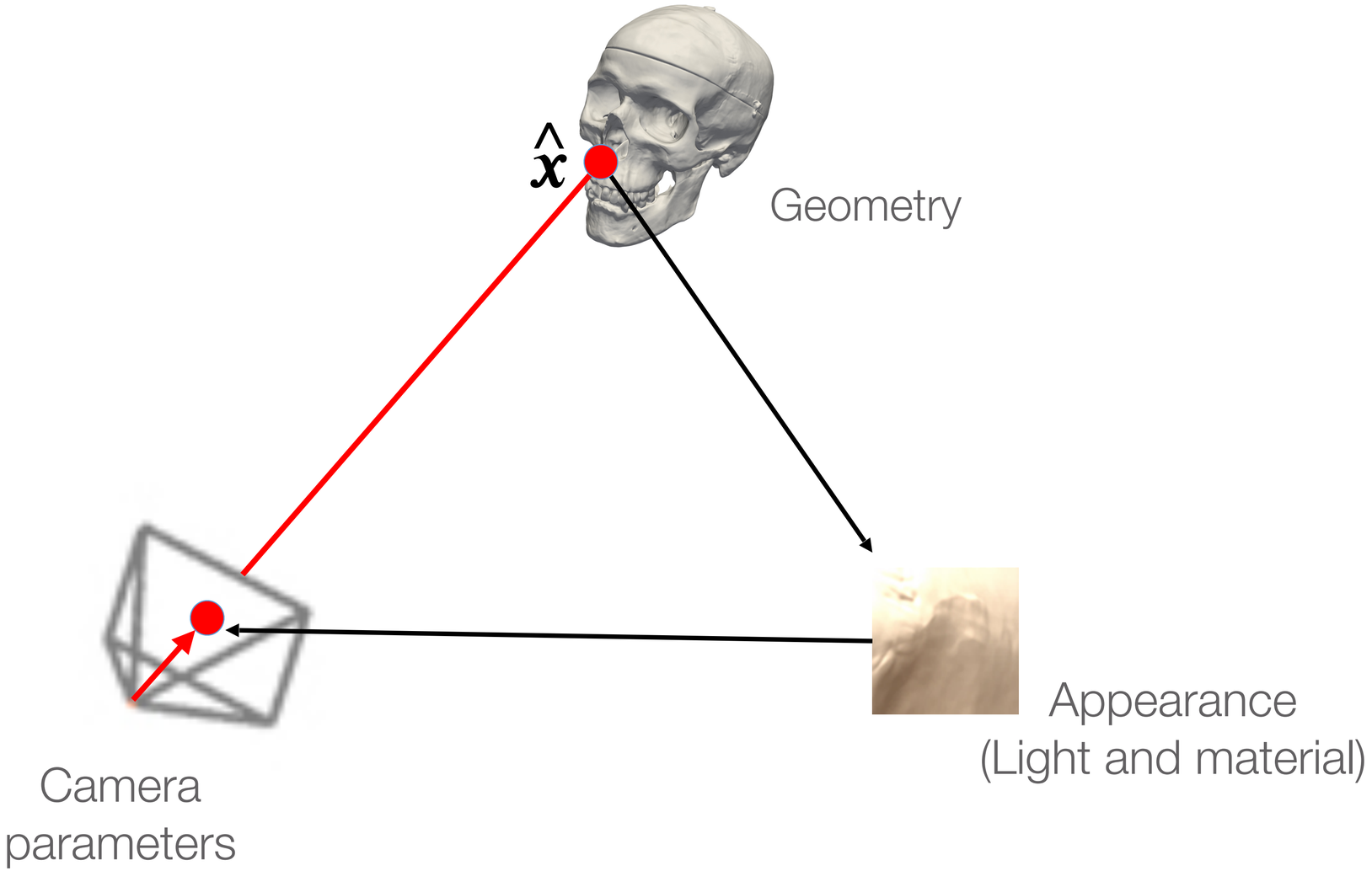
Volume density thresholds of NeRF



$\sigma = 1$ $\sigma = 50$ $\sigma = 500$

UNISURF: Unifying Neural Implicit Surfaces and Radiance Fields for Multi-View Reconstruction, Oechsle et al., 2022

Differential **Surface** Rendering



Implicit Differentiable Renderer (IDR)

Neural Geometry

Neural Appearance

Camera parameters

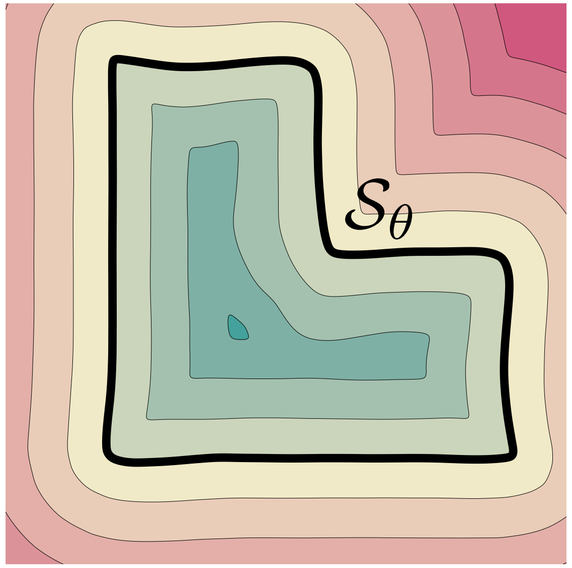
Differentiable Renderer

Input image

Loss

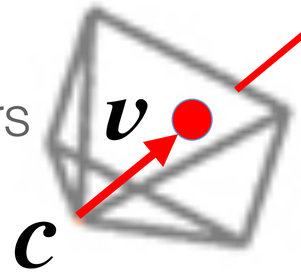
Multiview neural surface reconstruction by disentangling geometry and appearance (IDR), Yariv et. al., NeurIPS 2020

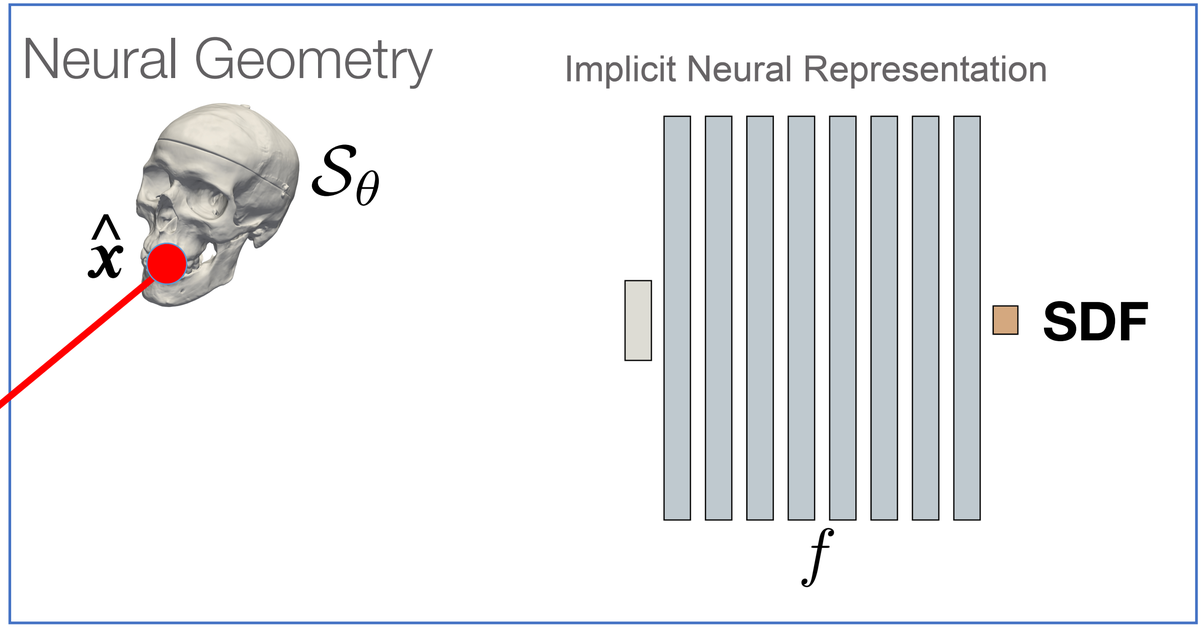
Neural Geometry



SDF

Camera parameters

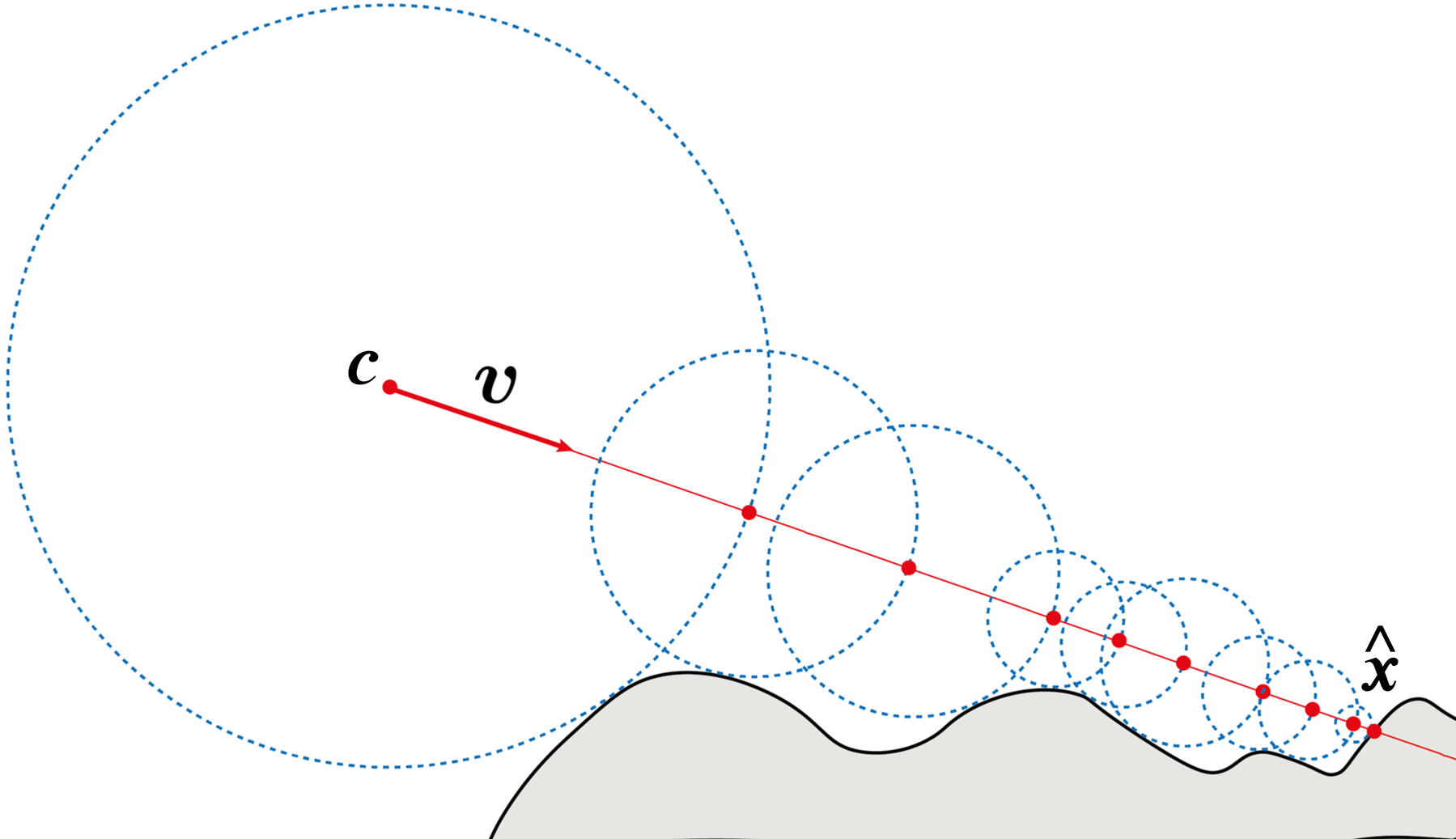




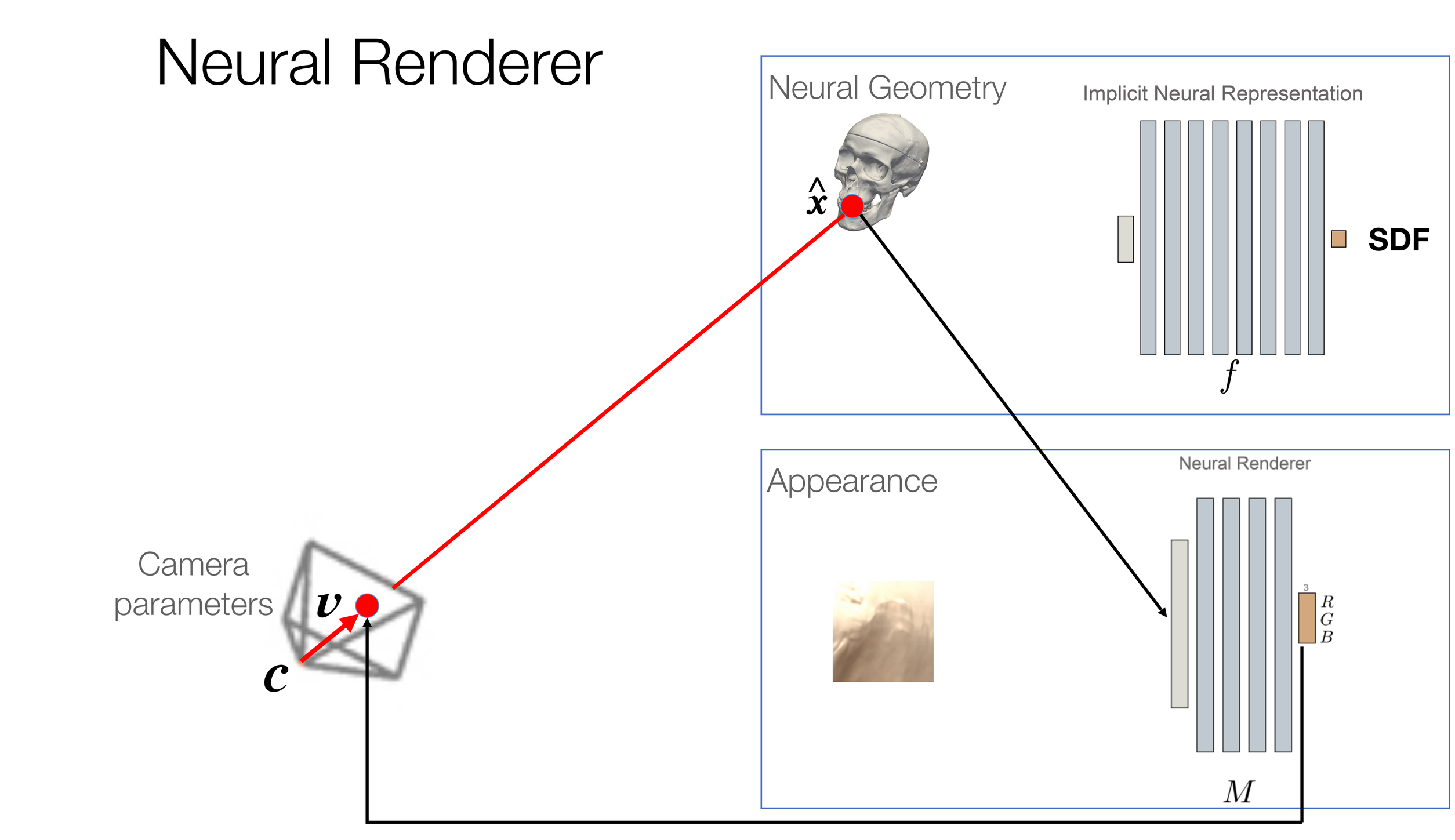
$$\mathcal{S}_\theta = \{\boldsymbol{x} \in \mathbb{R}^3 \mid f(\boldsymbol{x}; \theta) = 0\}$$

Sphere Tracing

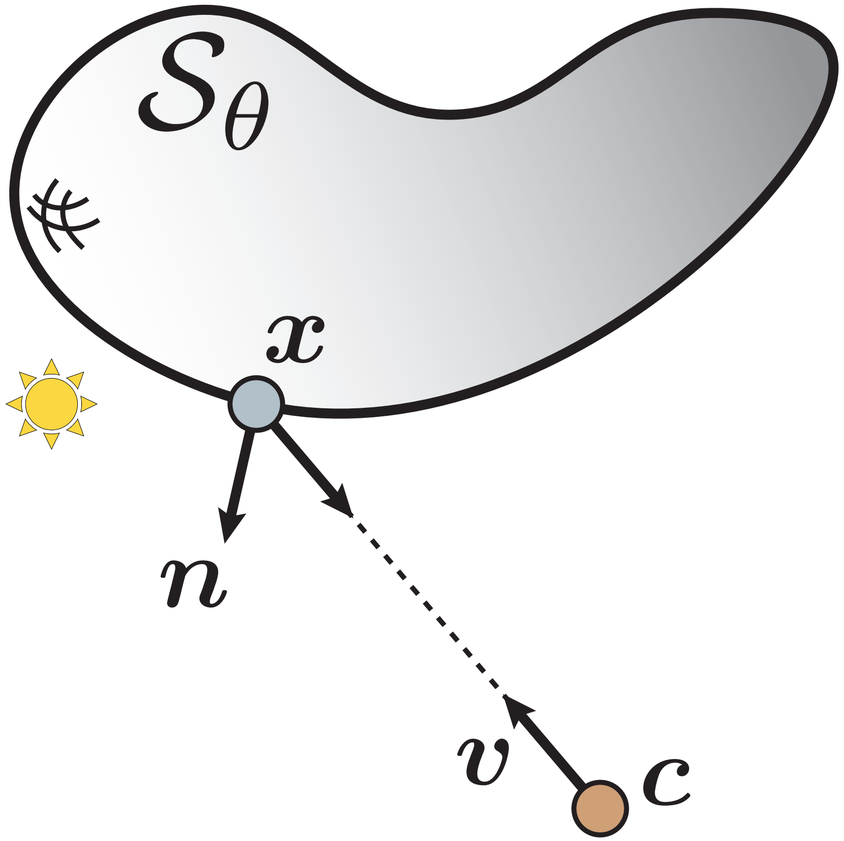
- Finding intersection point

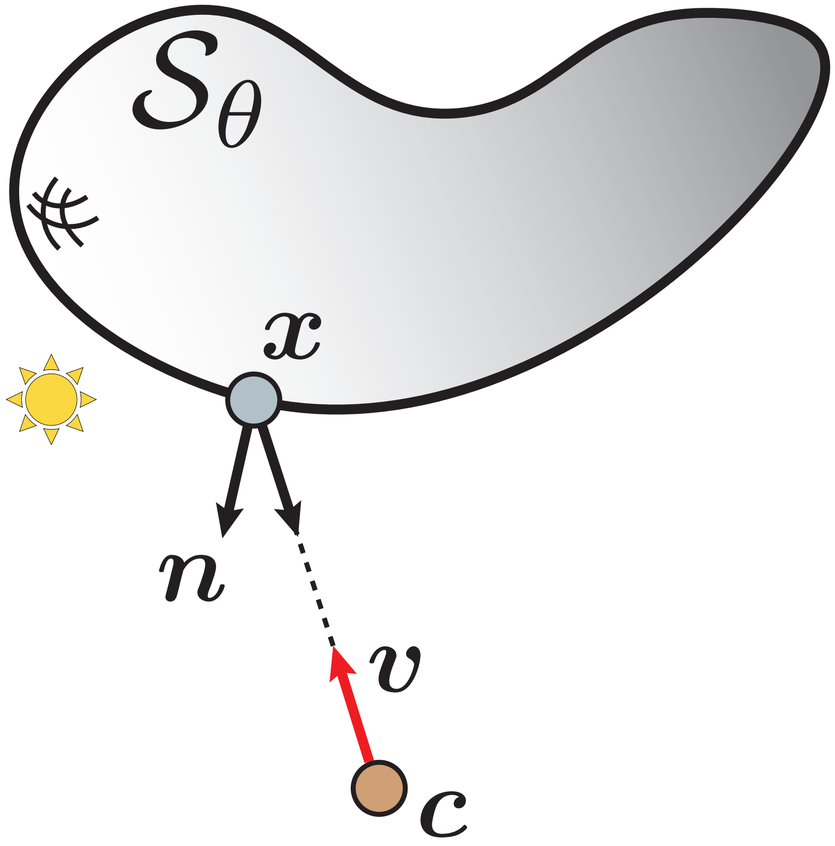


Neural Renderer



View dependent color





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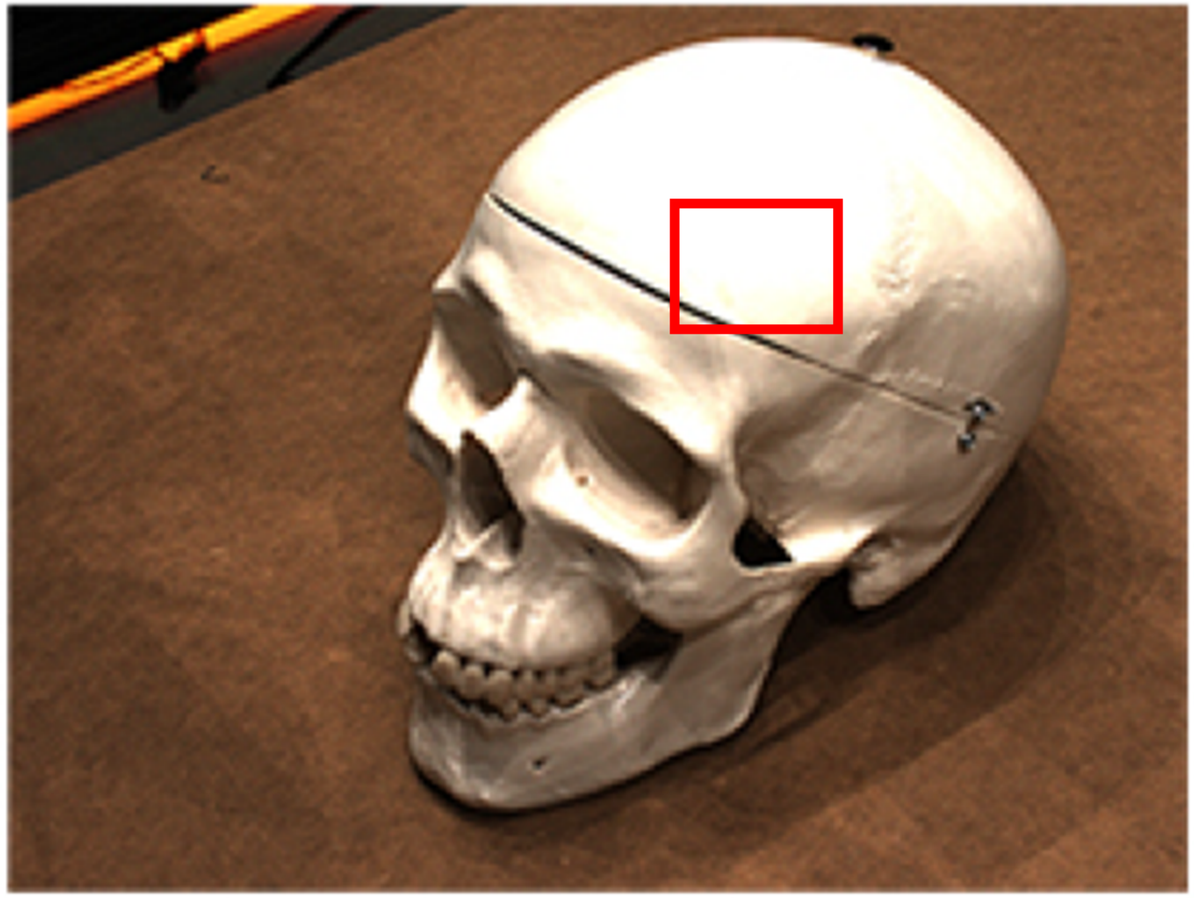
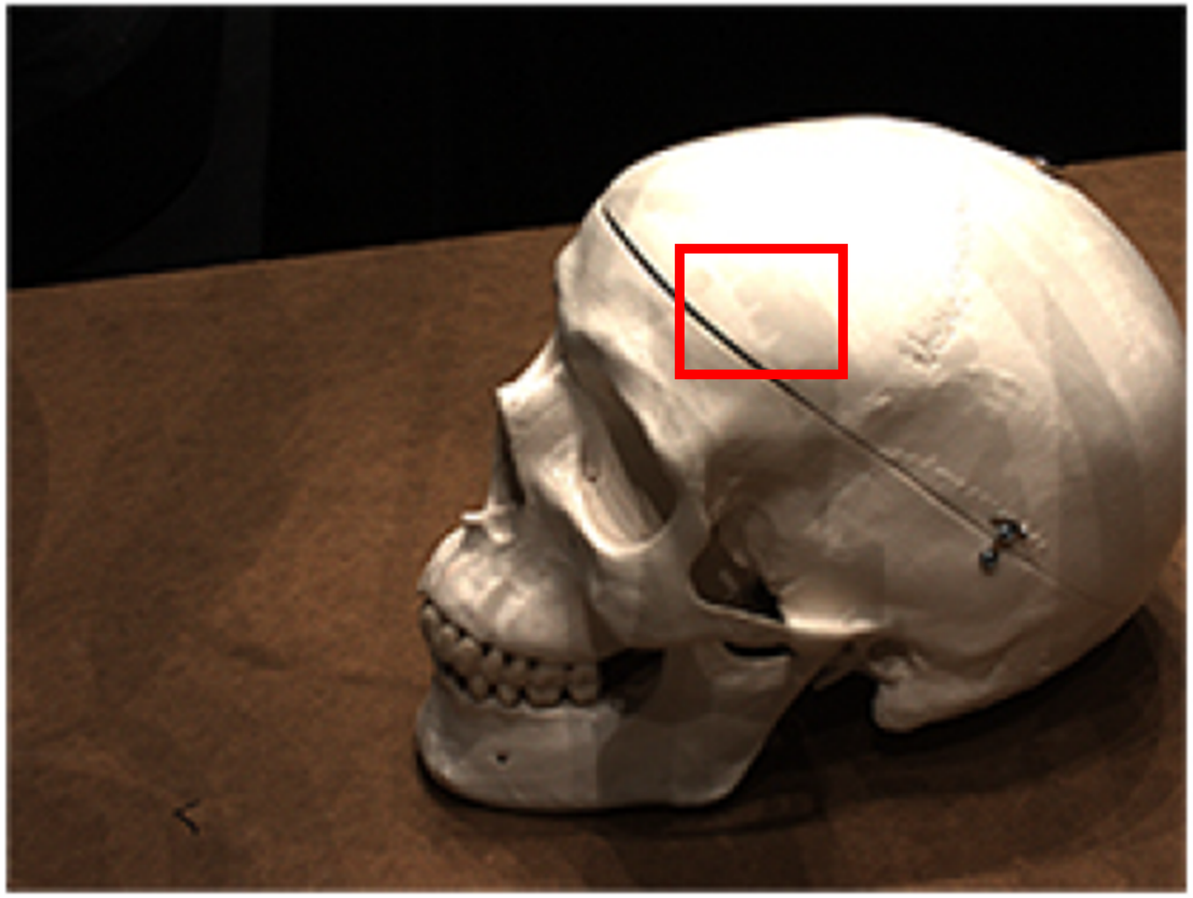
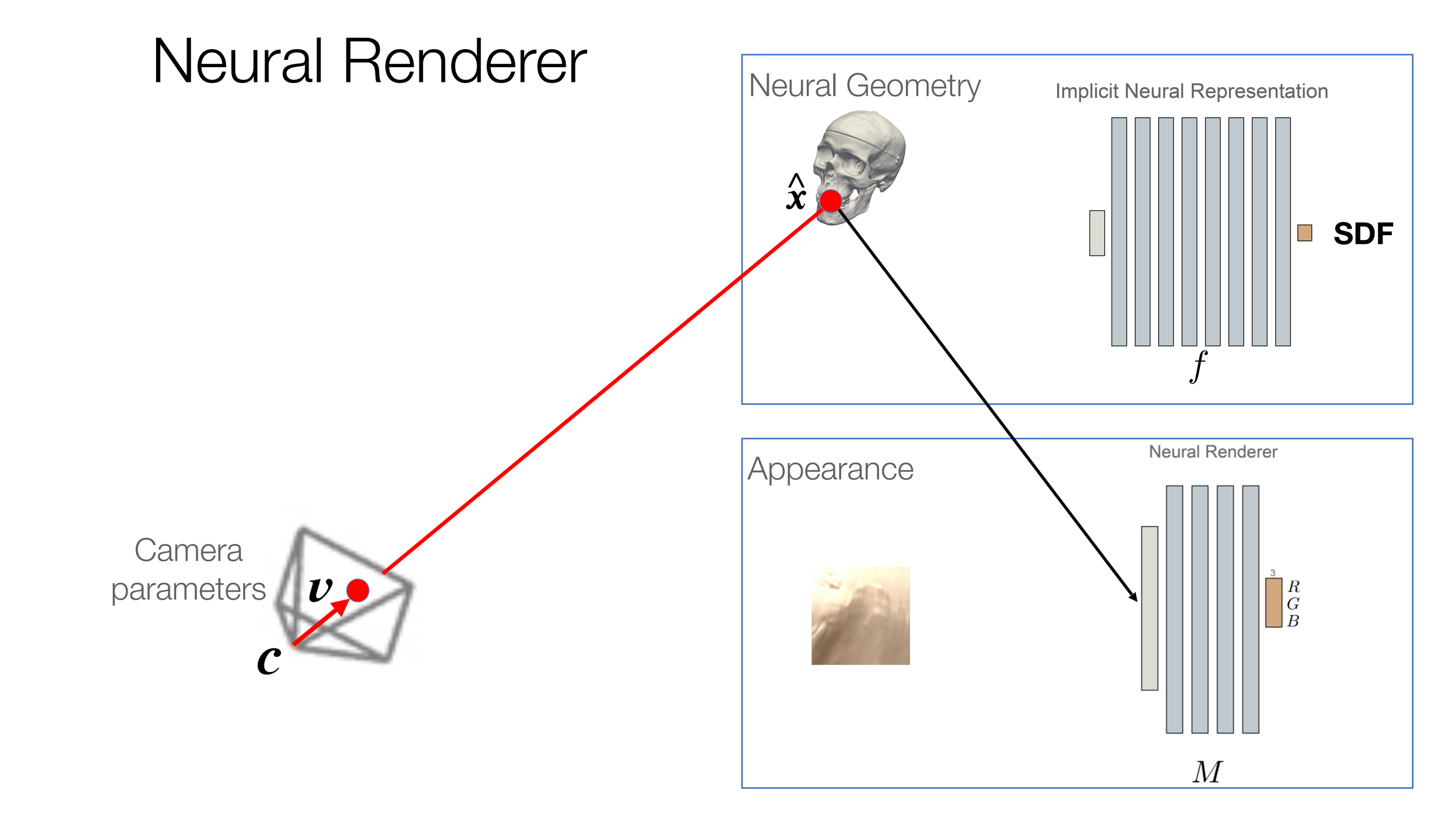
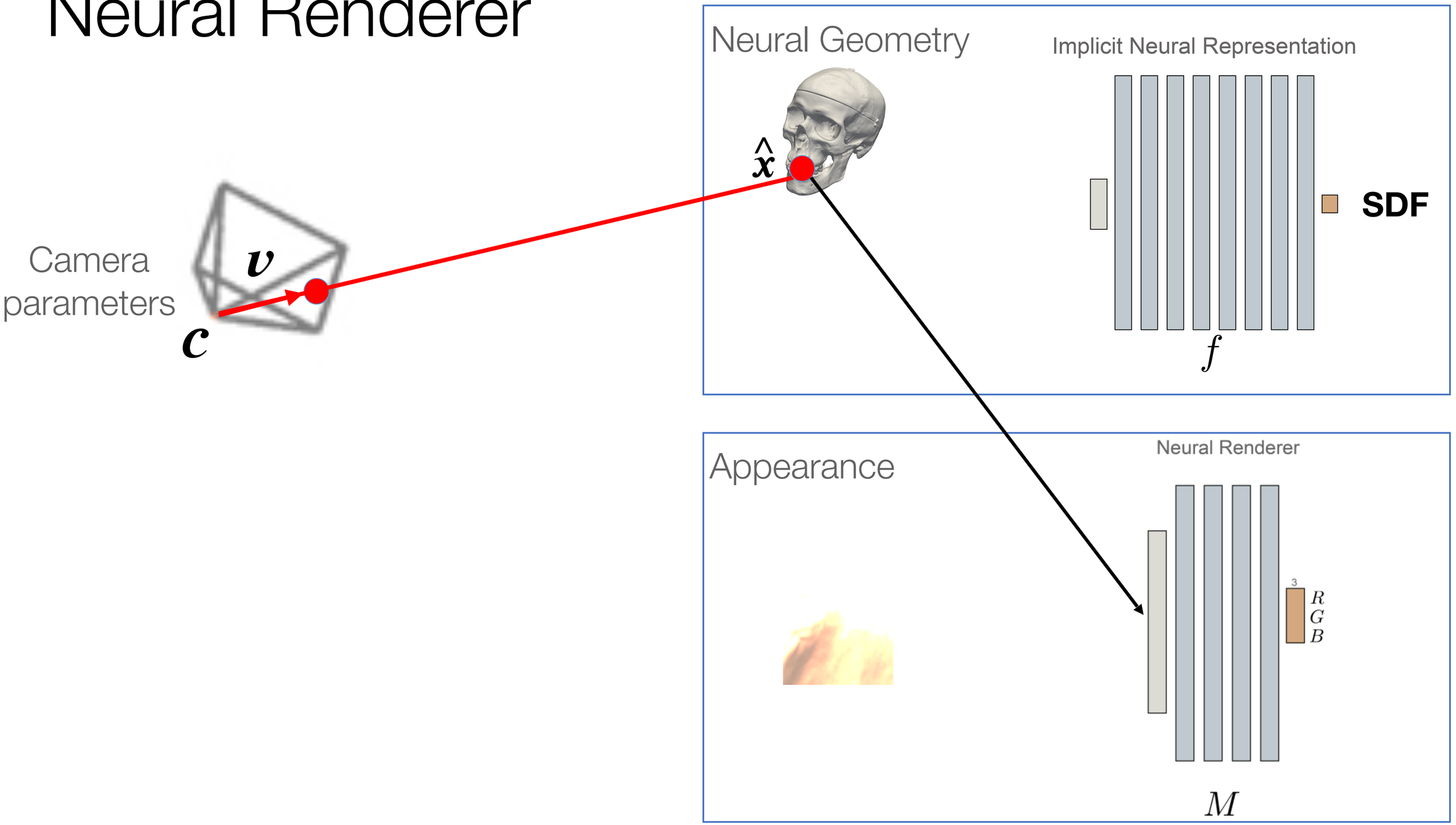




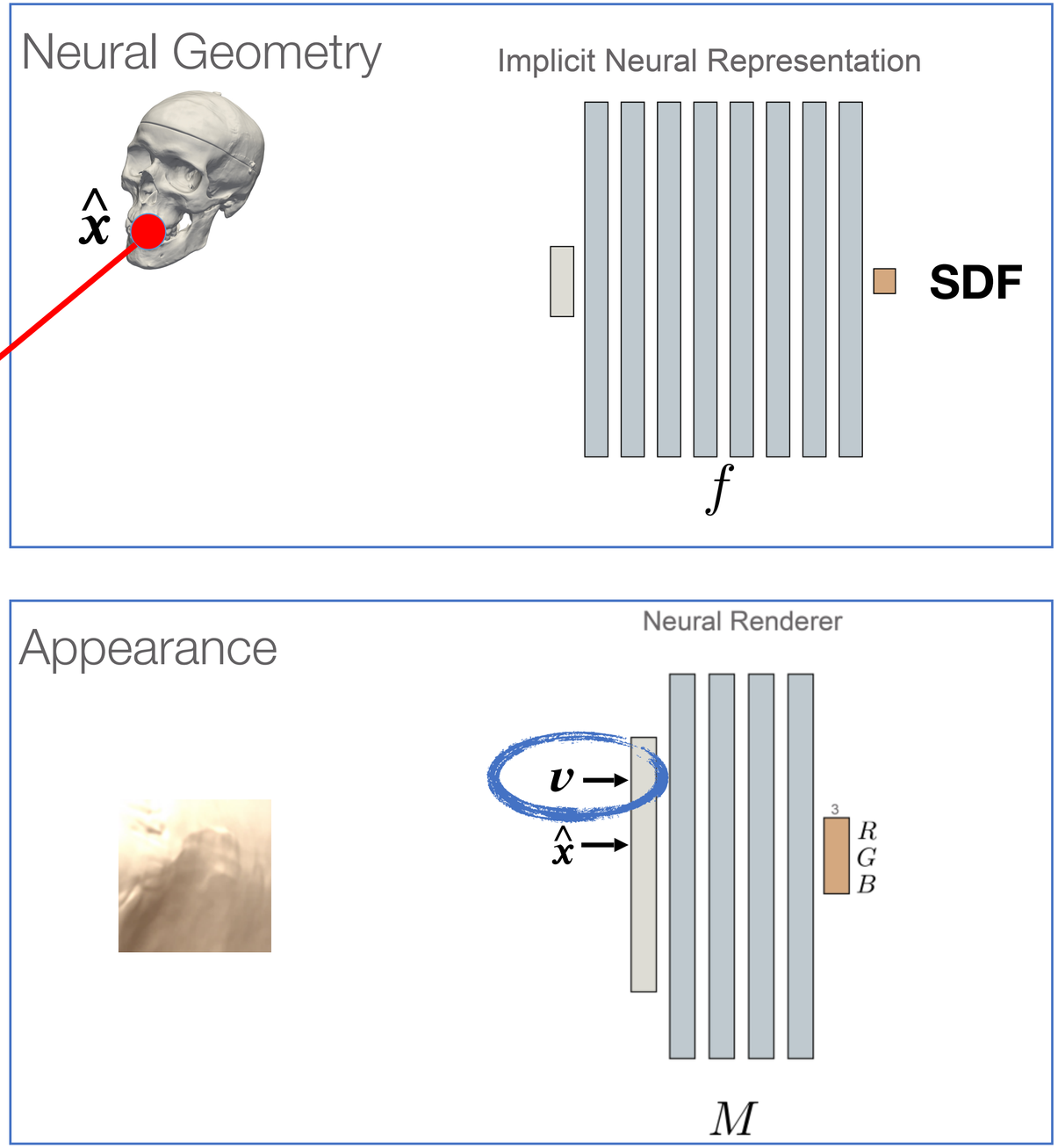
Neural Renderer

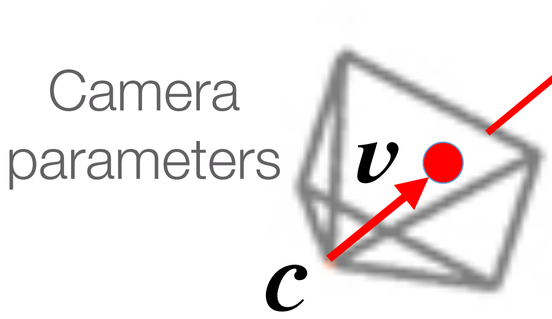


Neural Renderer



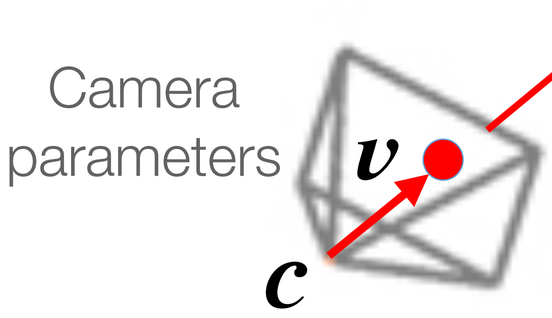
Neural Renderer

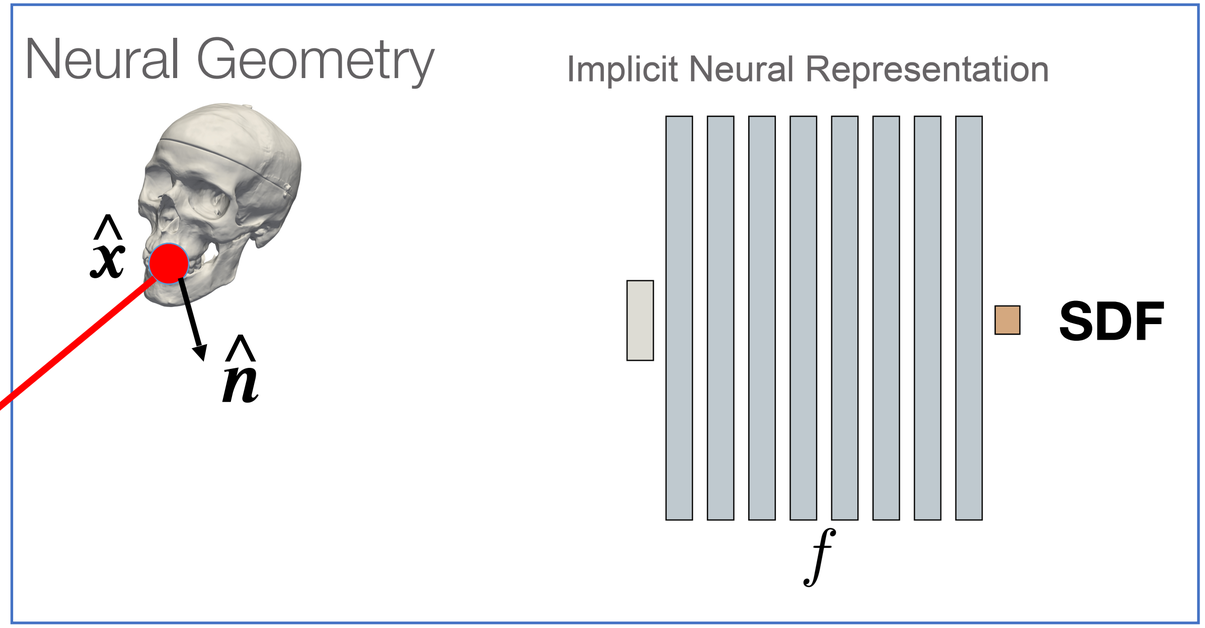


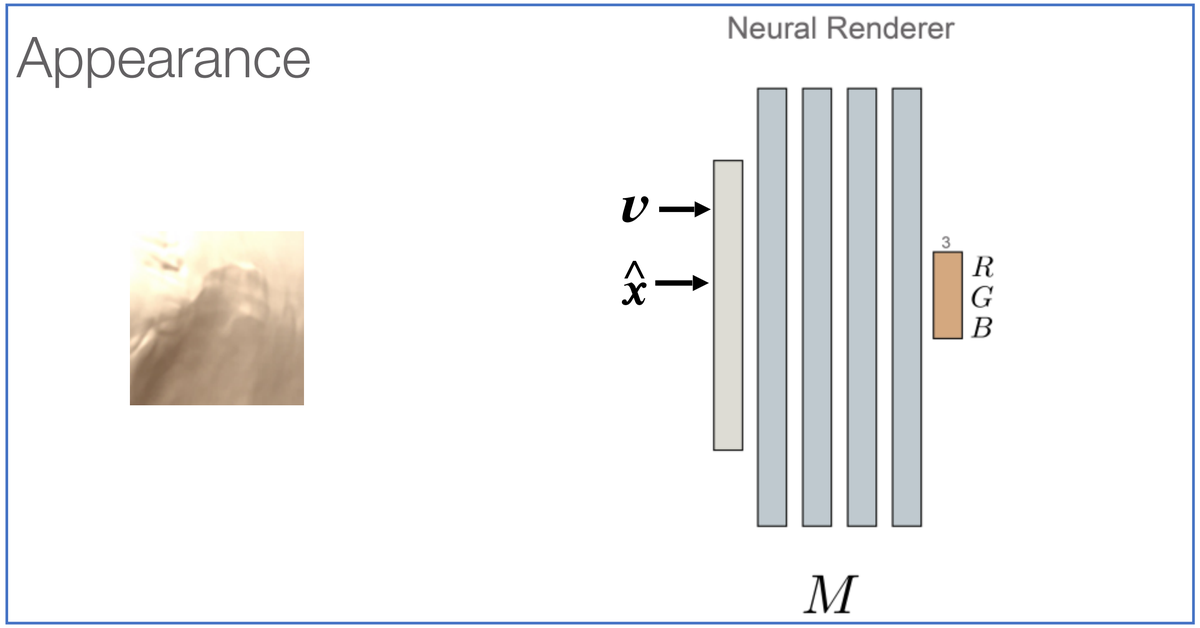


Neural Renderer

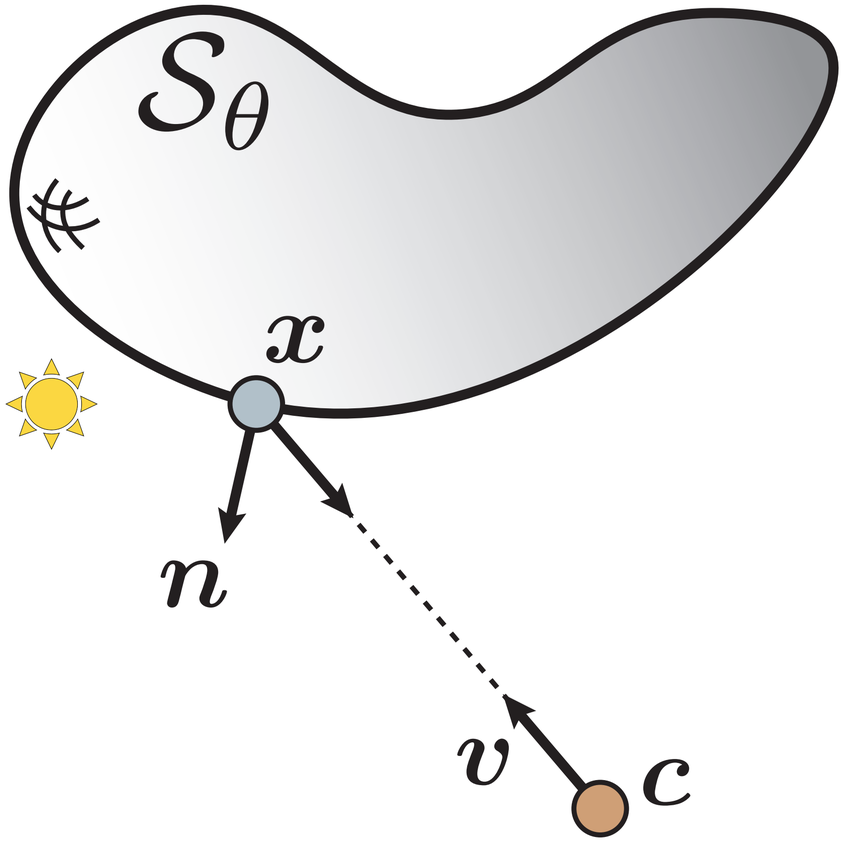
- Can we render a different geometry with the same renderer?
- What kind of input can “encourage” the renderer to generalize?

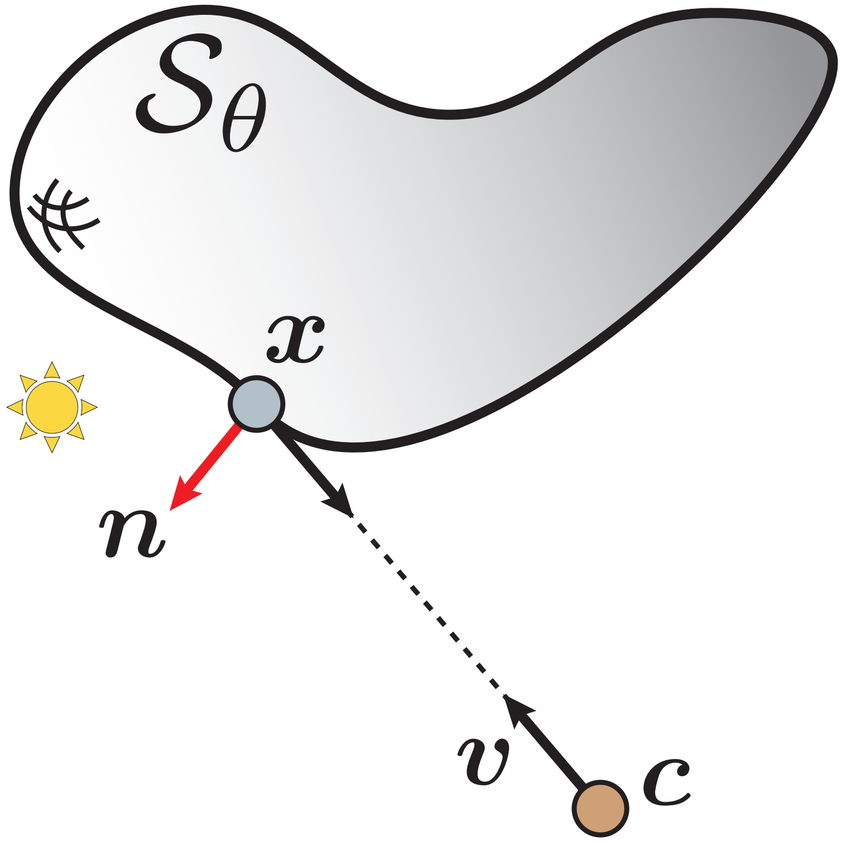






“Geometry” dependent color





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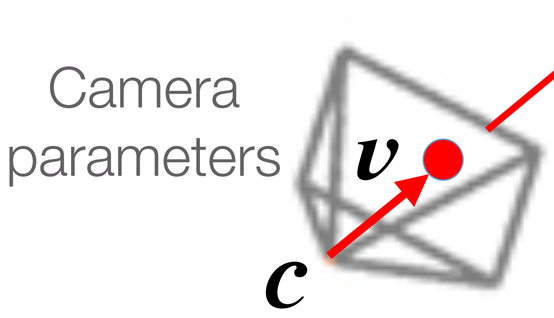
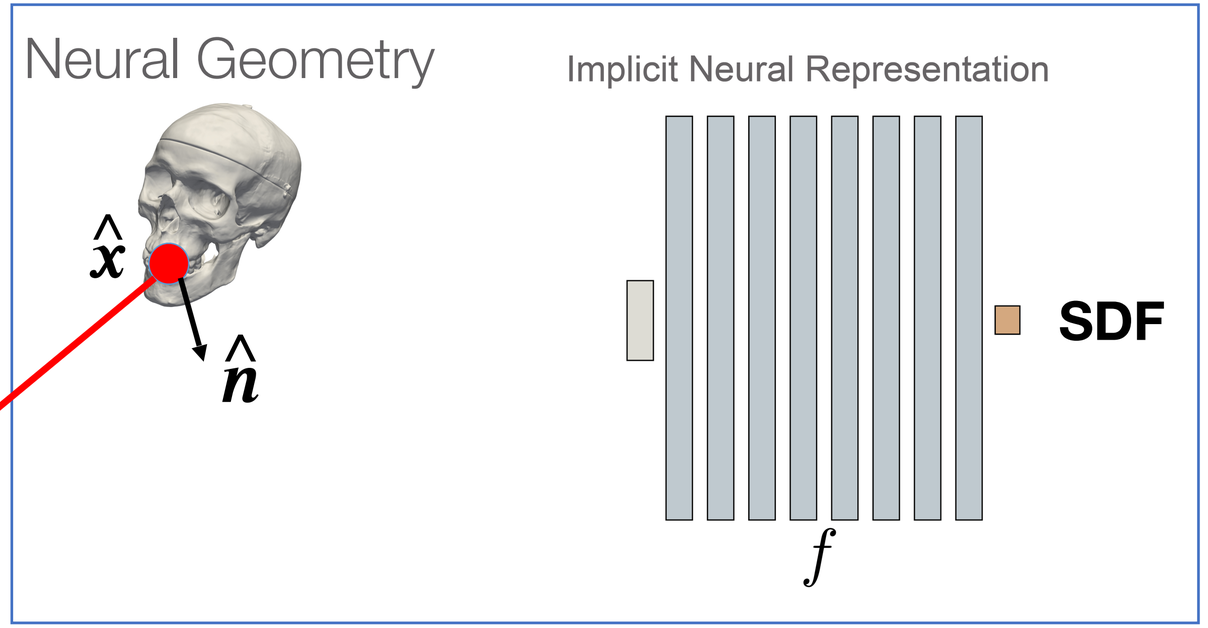
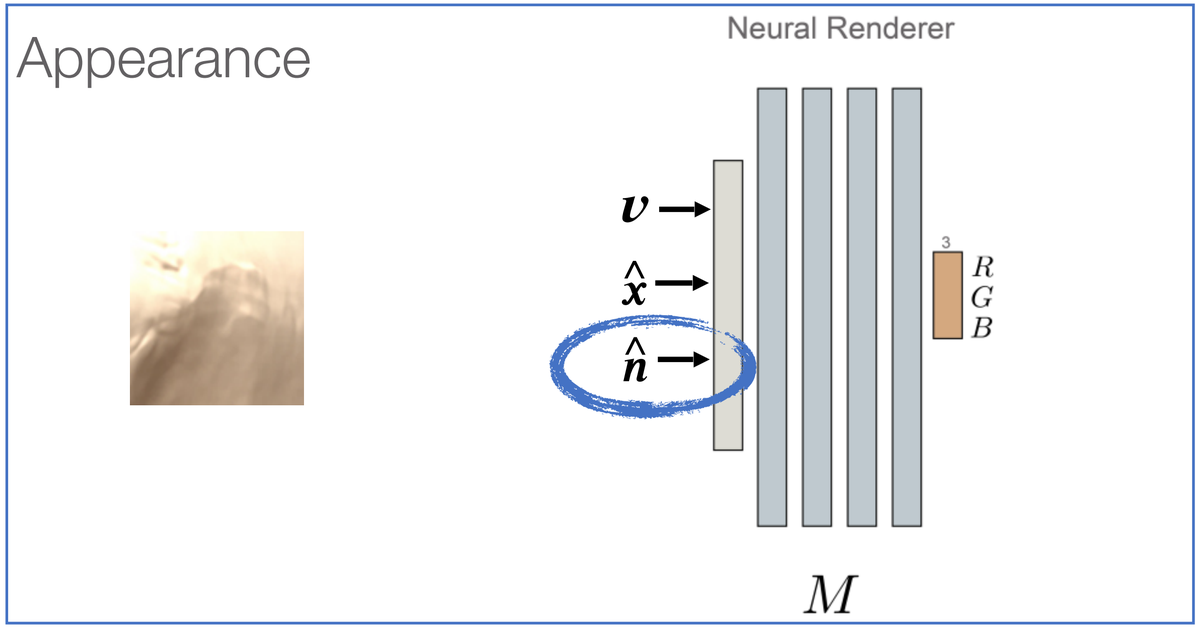


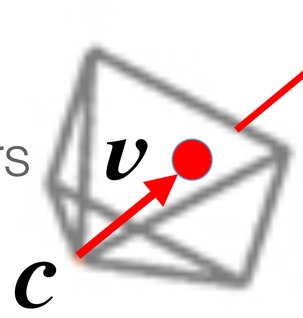




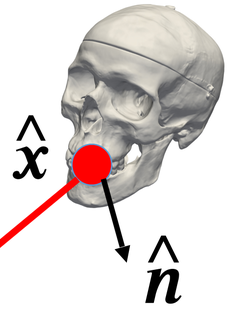
Neural Renderer

- Adding a global feature to allow secondary lighting effects and self shadows

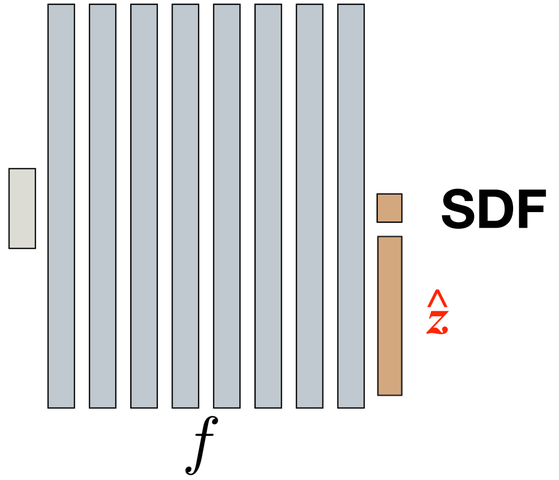
Camera parameters



Neural Geometry

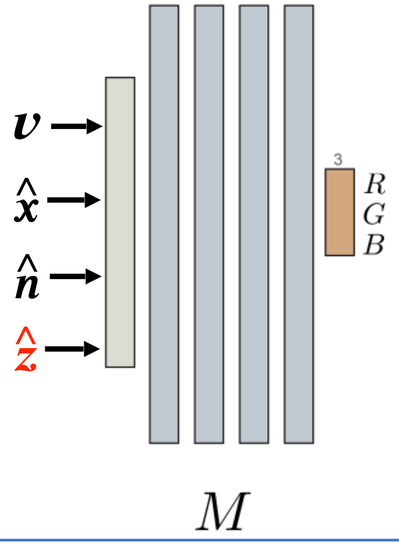


Implicit Neural Representation



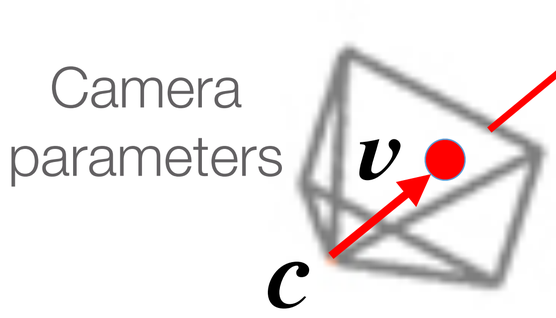
Appearance

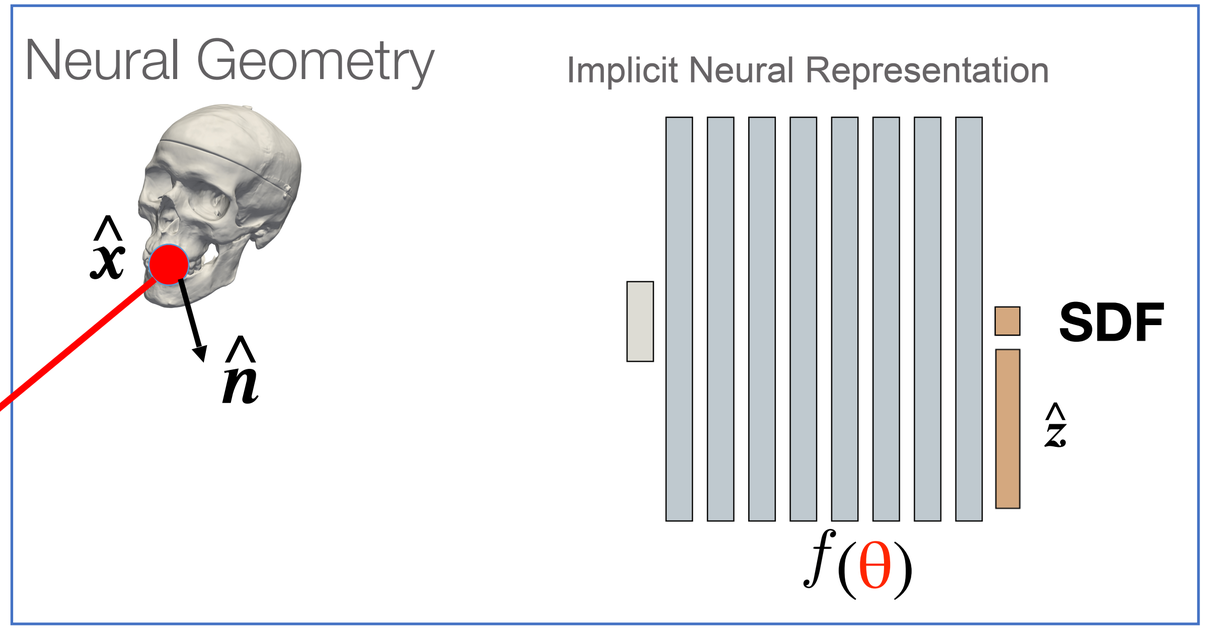
Neural Renderer

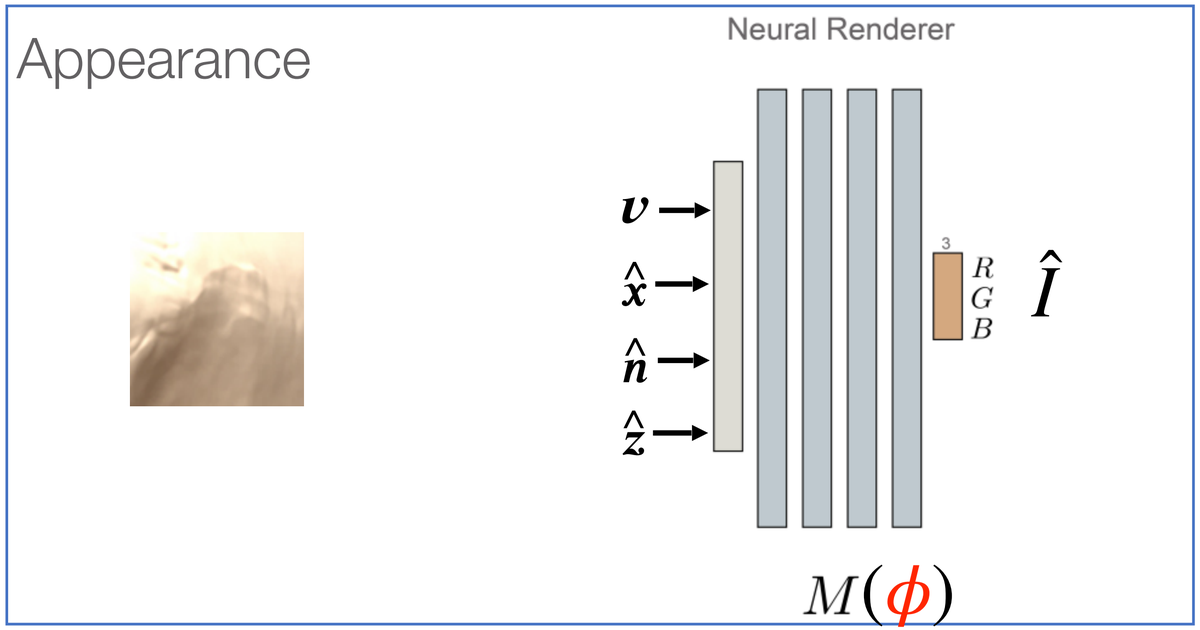


Training

- Loss: $|\hat{I} - I|$

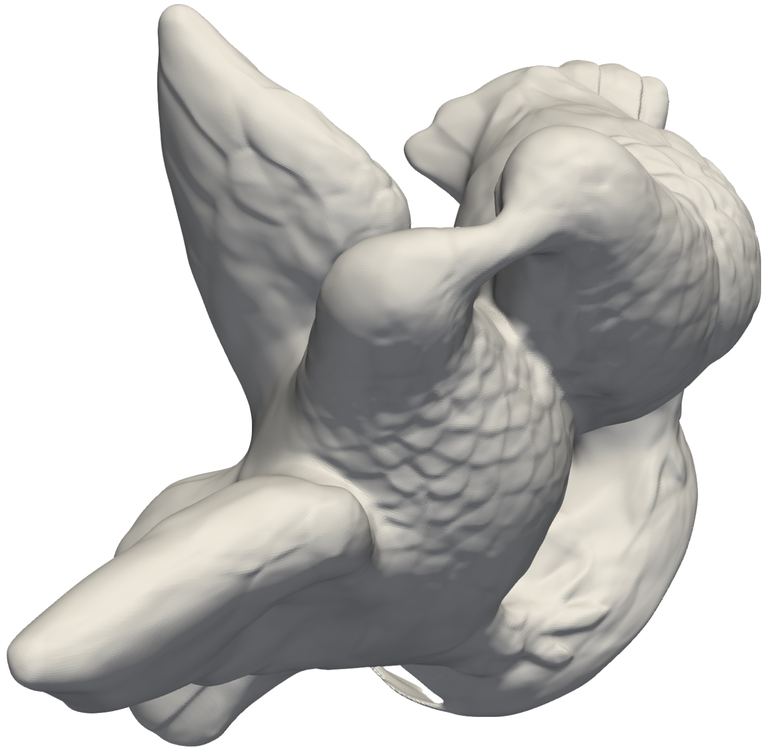




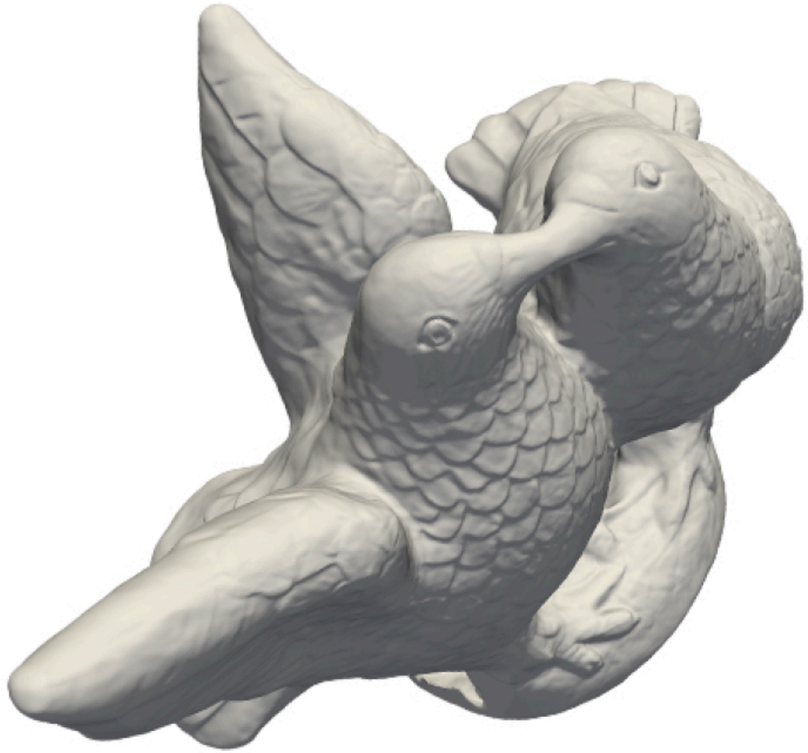


Positional encoding

$$\mathrm{PE}(\boldsymbol{y}) = \left(\sin(2^0\pi\boldsymbol{y}), \cos(2^0\pi\boldsymbol{y}), \cdots, \sin(2^{L-1}\pi\boldsymbol{y}), \cos(2^{L-1}\pi\boldsymbol{y})\right)$$



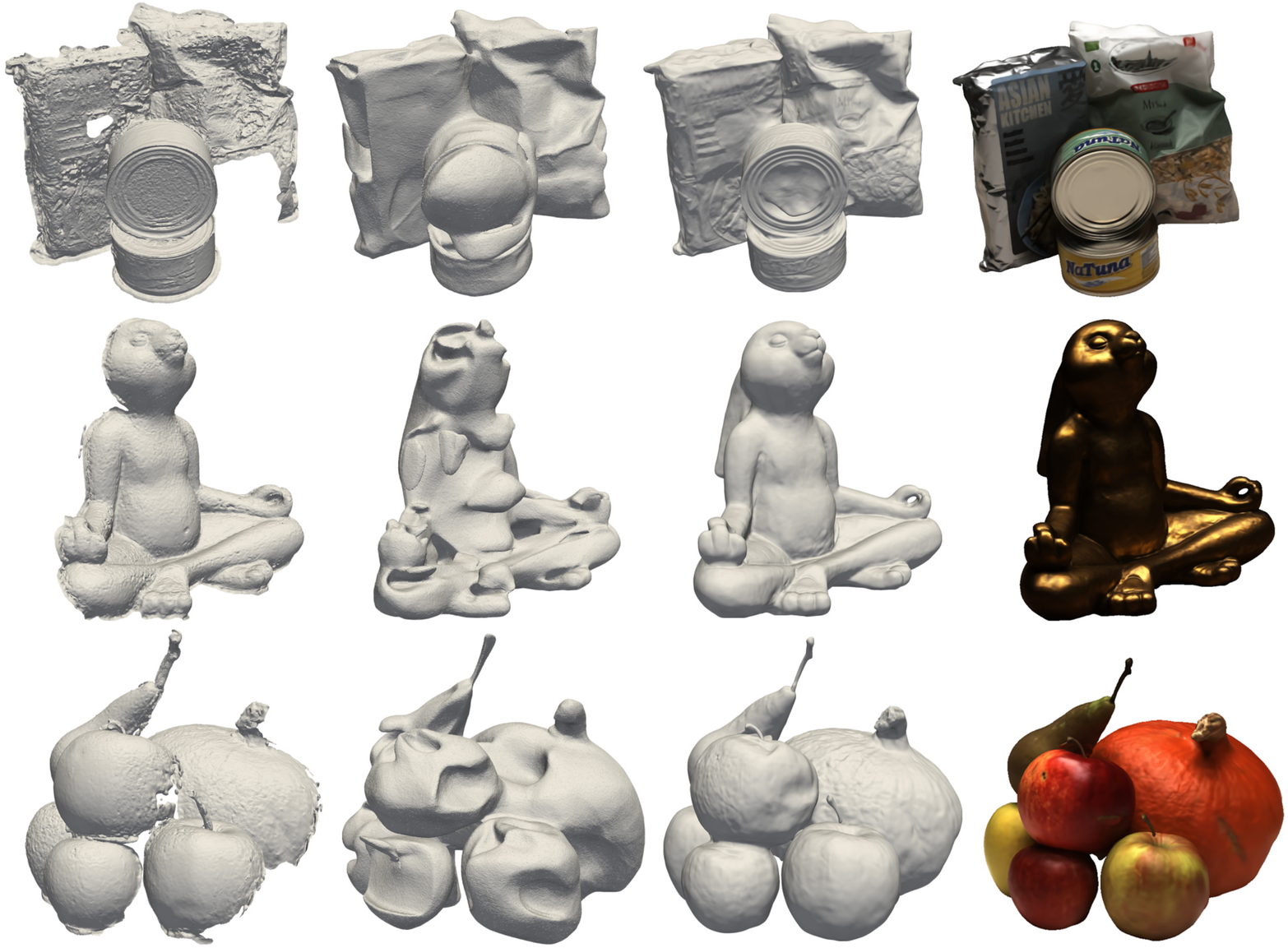
No PE, 5000 epochs



2000 epochs

NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis, Mildenhall, Srinivasan, Tancik, et al., 2020

Results: comparisons



ColMap + sPSR | DVR $(\hat{x}, \hat{n}, v)$ | IDR | IDR - rendering

Results: ablation study

Input images

$M(\hat{\boldsymbol{x}}, \hat{\boldsymbol{n}}, \boldsymbol{v}, \boldsymbol{z})$

$M(\hat{\boldsymbol{x}}, \hat{\boldsymbol{n}}, \boldsymbol{v}, \boldsymbol{z})$

$M(\hat{\boldsymbol{x}}, \hat{\boldsymbol{n}}, \boldsymbol{v}, \boldsymbol{z})$

$M(\hat{\boldsymbol{x}}, \hat{\boldsymbol{n}}, \boldsymbol{v}, \boldsymbol{z})$

Surface Rendering v.s. Volume Rendering

Surface Rendering:

- Representation: Implicit surface
- Rendering: Find intersection
- Back-propagate: 1 sample

DVR [Niemeyer et al. ’20]

IDR [Yariv et al. ’20]

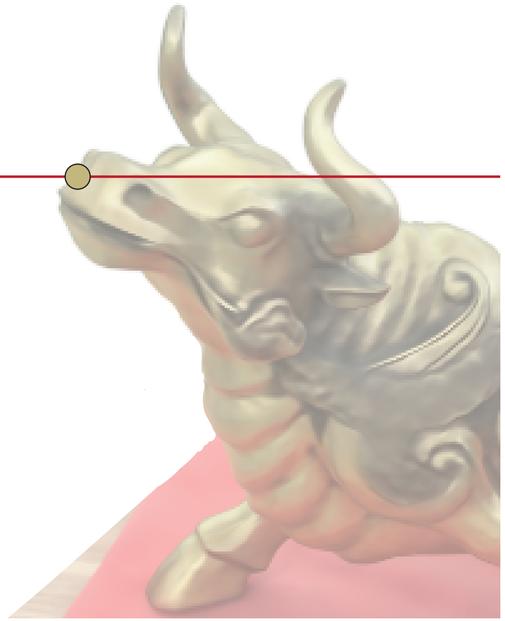
- Limitation:

  Object masks

IDR [Yariv et al. ’20]

No Masks

With Masks



Volume Rendering:

- Representation: Volume density
- Rendering: Integral approximation
- Back-propagate: multiple samples

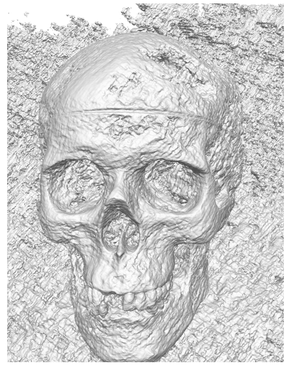
NeRF [Mildenhall et al. ’20]

…

- Limitation:

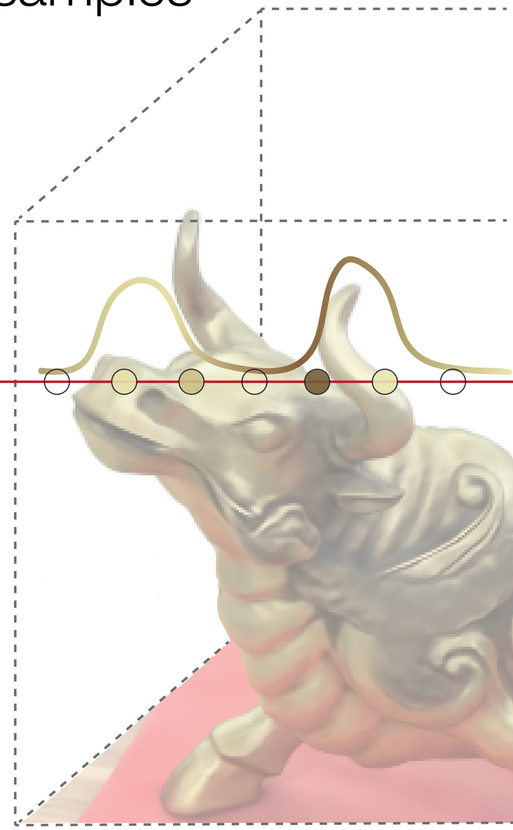
  Noisy geometry

NeRF [Midenall et al. ’20]



Density = 50

Rendering



Surface Rendering v.s. Volume Rendering

Surface Rendering:

- Representation: Implicit surface
- Rend
- Back
  - DVR [Ni
  - IDR [Yar
- Limita
  Objec

Volume Rendering:

- Representation: Volume density
- s

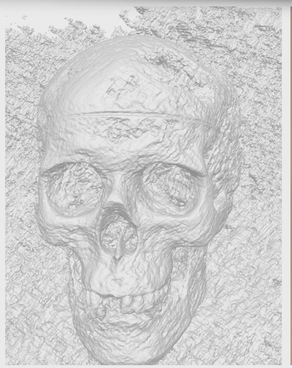
Can we get the best of both worlds?

IDR [Yariv et al. '20]

No Masks

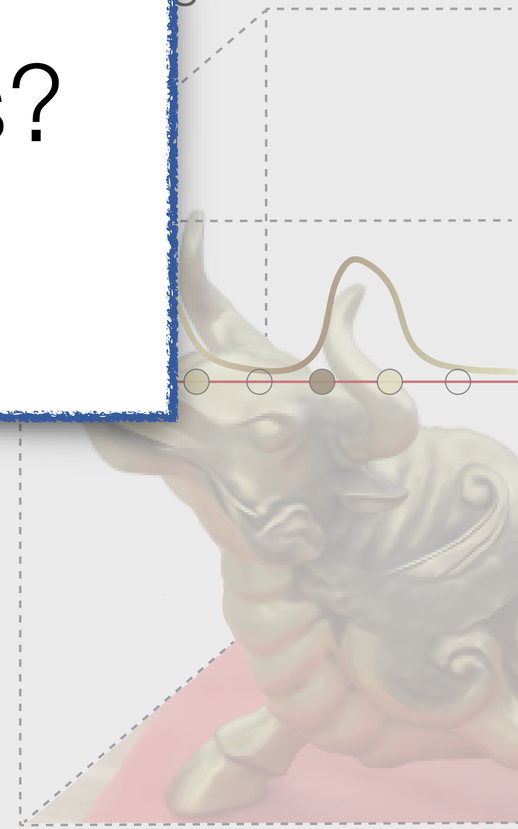
With Masks

NeRF [Midenall et al. '20]



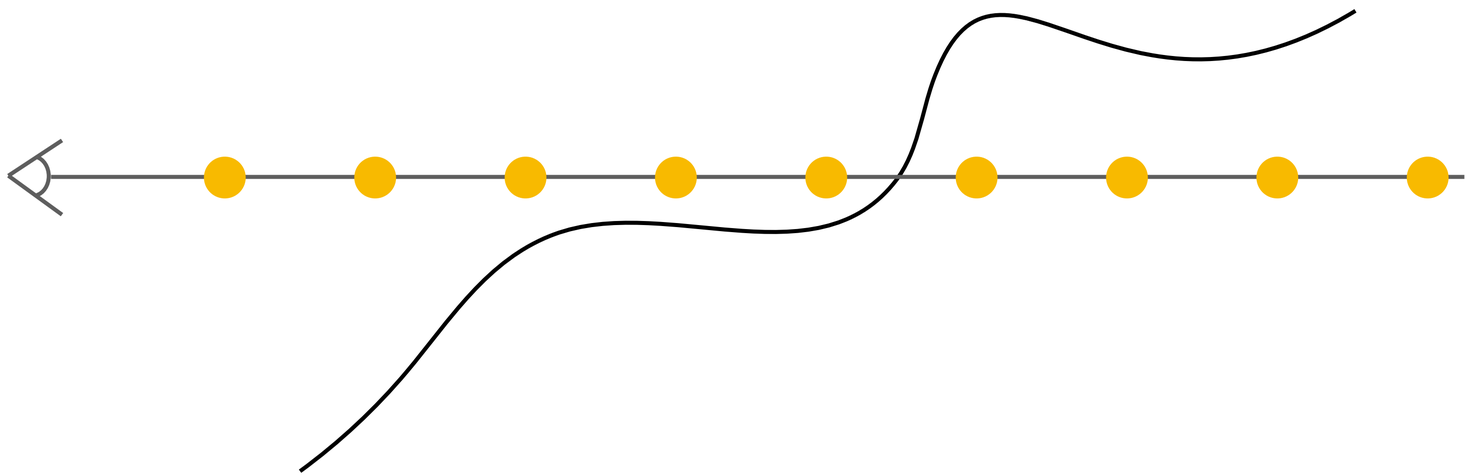
Density = 50

Rendering



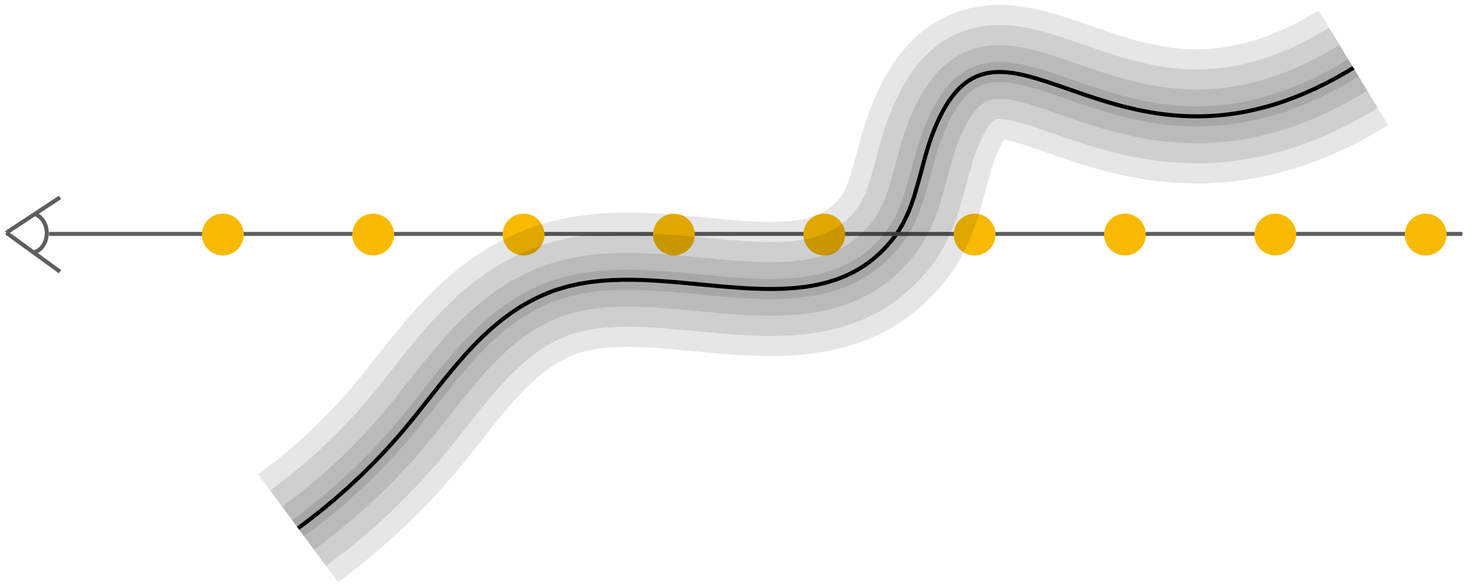
Surface reconstruction using volume rendering

How can we volume-render a surface?



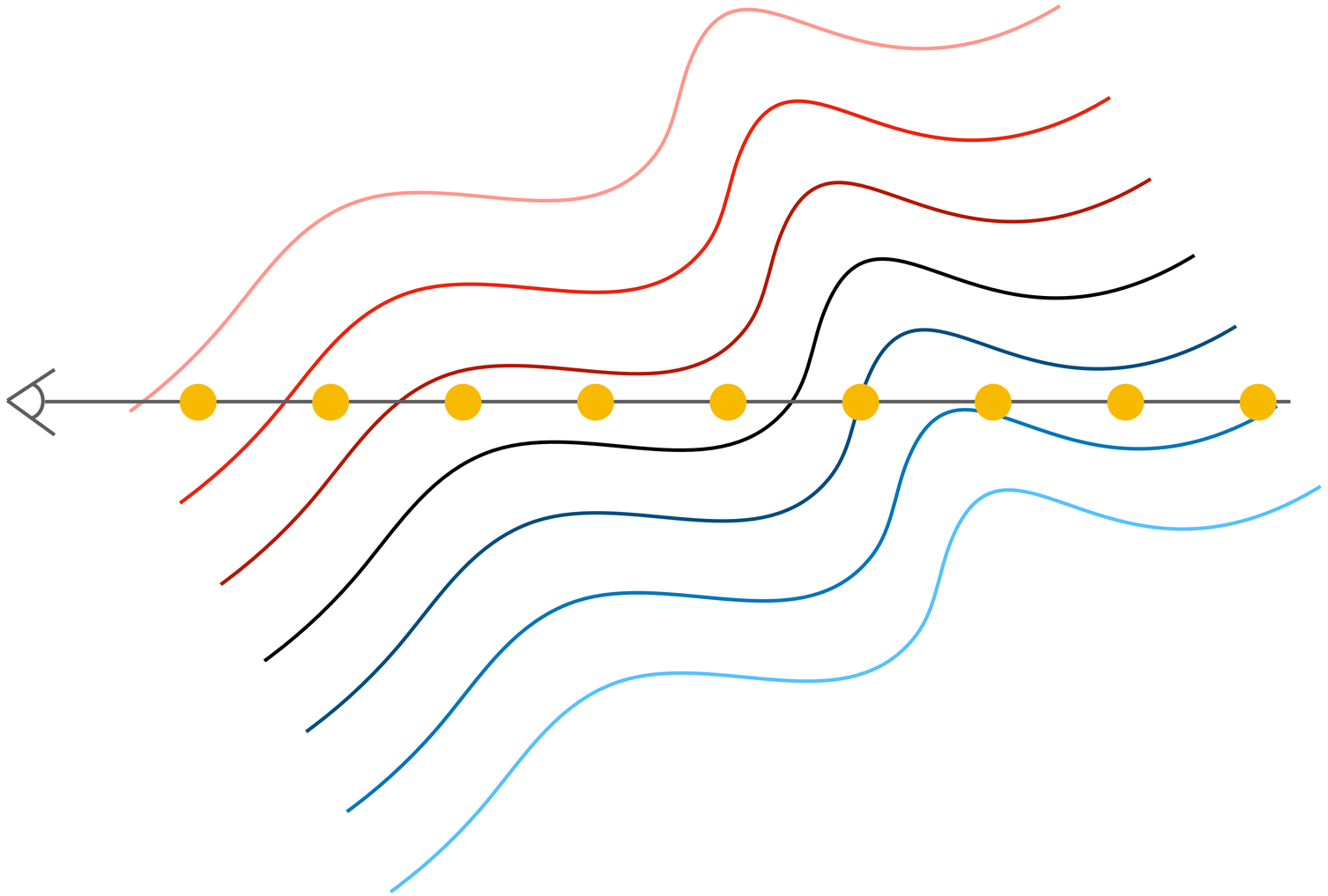
Surface reconstruction using volume rendering

⇒ Represent the scene as a “soft” surface



Surface reconstruction using volume rendering

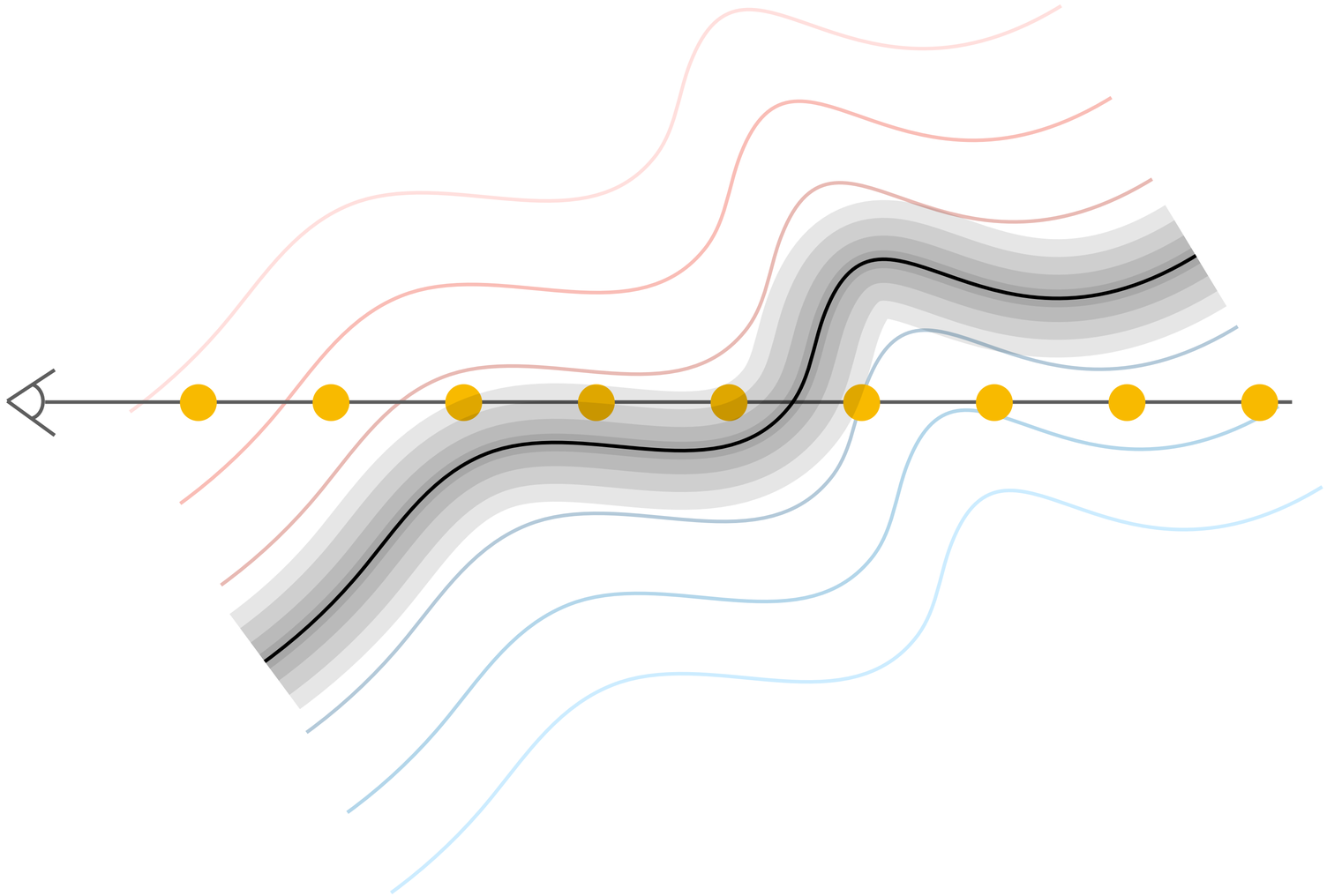
MLP-parameterized signed distance function $f(\mathbf{x})$



Volume rendering of neural implicit surfaces, Yariv et al., NeurIPS 2021

Surface reconstruction using volume rendering

Modeling density as: $\sigma(\mathbf{x}) = \alpha \Psi_\beta(f(\mathbf{x}))$

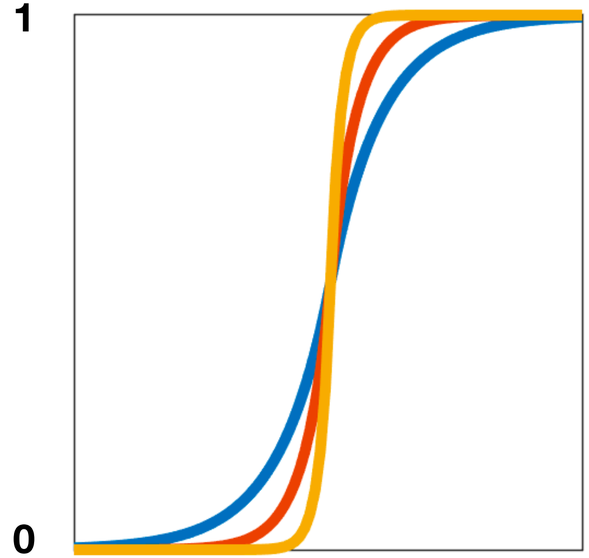


Volume rendering of neural implicit surfaces, Yariv et al., NeurIPS 2021

VolSDF: density σ as transformed SDF

Modeling density as: $\sigma(\mathbf{x}) = \alpha \Psi_\beta(f(\mathbf{x}))$

Density



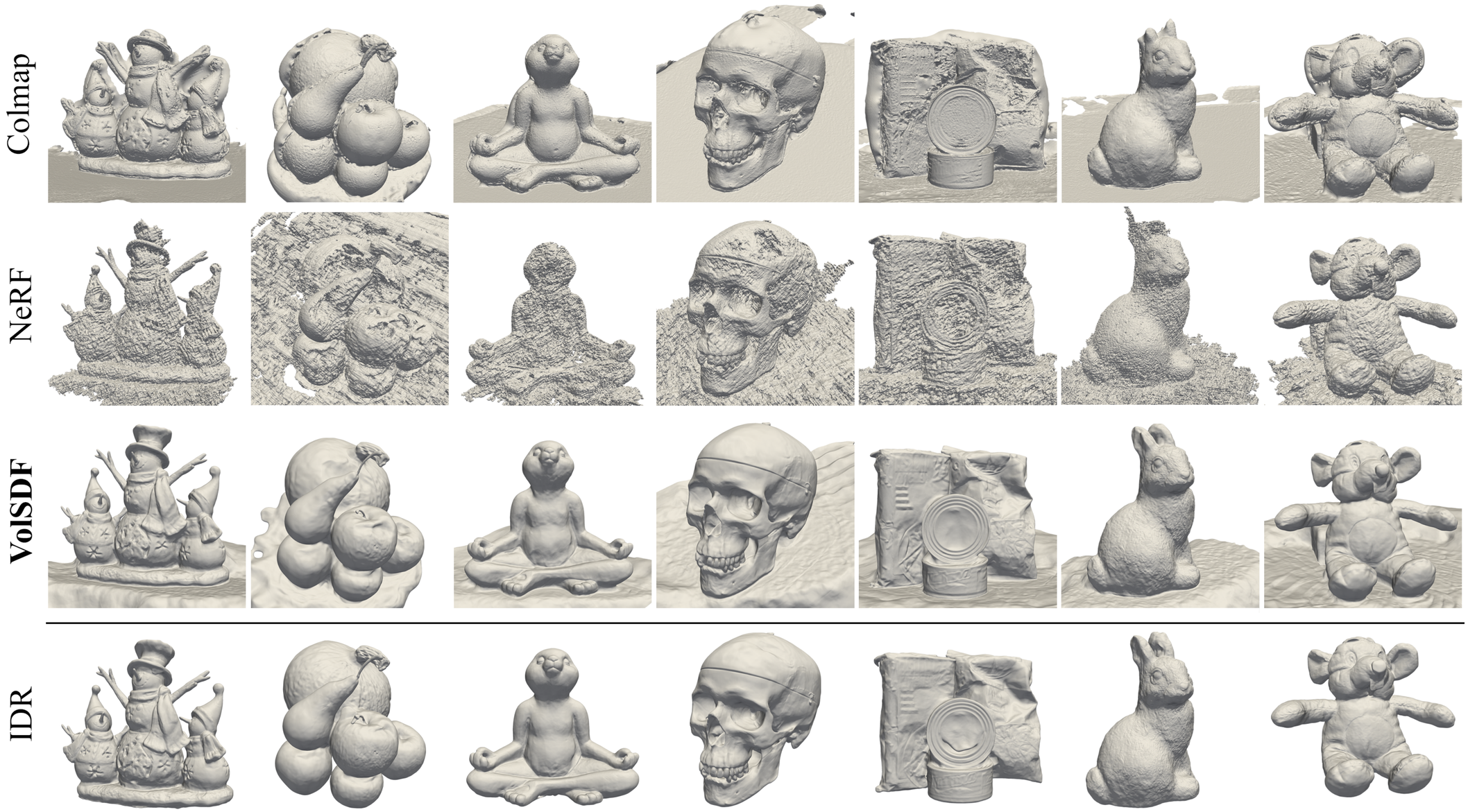
Laplace CDF

Signed Distance Function

*<u>Recall:</u> NeRF models Density $\sigma(\boldsymbol{x}) : \mathbb{R}^3 \rightarrow \mathbb{R}^+$ is a general purpose MLP

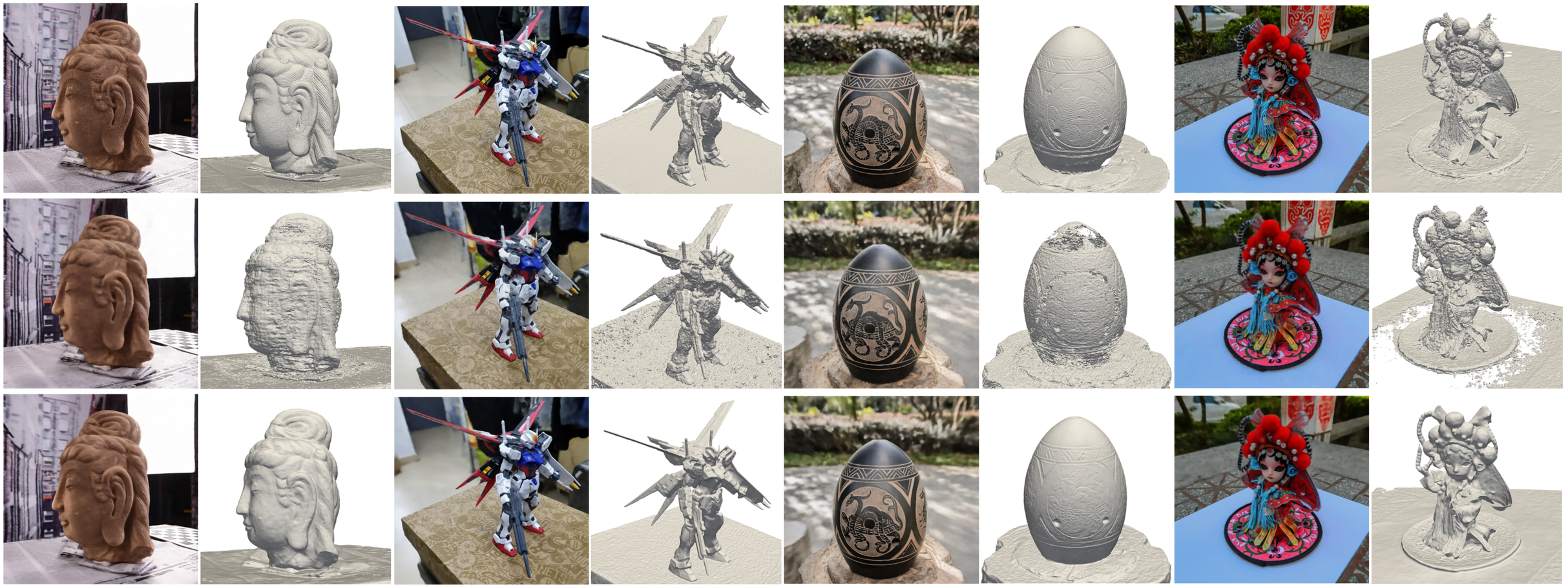
Volume rendering of neural implicit surfaces, Yariv et al., NeurIPS 2021

Results: comparisons



DTU: Large Scale Multi-view Stereopsis Evaluation [Jensen et al. 2014]

Results: comparisons



BlendedMVS: A Large-scale Dataset for Generalized Multi-view Stereo Networks [Yao et al. 2020]

Summary

Take-away messages:

- The underlying geometry, obtained by volume renderers is non-smooth and contains artifacts.
- Differentiable surface renderers produce highly accurate 3D reconstructions.
  However, they depend on object masks.
- Unifying surface and volume rendering is possible!

Limitation:

- Per-scene overfit (no generalization)
- Computational expensive
- Assuming fully opaque surface

Other applications:

- 3D Generative models
- Dynamic scenes
- Camera optimization
- Material-Light decomposition
- So many more …

Survey papers/blogs:

- NeRF Explosion
  [Frank Dellaert ’20]
- State of the Art on Neural Rendering
  [Tewari et al. '20]
- Advances in Neural Rendering
  [Tewari et al. ’21]
- Neural Fields in Visual Computing and Beyond
  [Xie at al. ’21]

Questions?